

Agricultural shocks and the dynamics of conflict risk: Evidence from desert locust swarms

Pierre E. Biscaye[†]

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Abstract

Can transitory economic shocks affect the long-term dynamics of violent conflict? Using data on conflict events and desert locust swarm exposure across 0.25° grid cells in Africa and the Arabian Peninsula from 1997-2018 in a staggered event study framework, I show that swarm exposure increases subsequent annual conflict risk by 1.8 percentage points (64%). Effects are consistent with initial agricultural destruction, as exposure in non-agricultural areas or outside the main crop growing season has no detectable impact. Yet conflict risk does not increase when the shock occurs, contrary to the prediction of a standalone opportunity cost mechanism. Instead, effects emerge later and are concentrated under broader instability, consistent with exposure amplifying conflict incidence rather than directly causing conflict onset. Grid-cell economic outcomes provide limited support for persistent opportunity cost effects, suggesting other channels may also matter for long-term impacts. Similar delayed and instability-contingent patterns following drought exposure suggest that these dynamics are not specific to locust swarms. Severe agricultural shocks can shape future conflict risk even when they do not generate immediate violence, with implications for how researchers estimate shock effects and how policymakers target post-shock recovery.

JEL codes: D74; O13; Q54; Q10

Keywords: conflict; agricultural shocks; desert locusts; natural disasters; long-term effects; Africa

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1 Introduction

A large economic literature studies how transitory agricultural shocks, such as weather and commodity price fluctuations, affect the risk of violent conflict. This is an important policy concern given the prominence of agricultural livelihoods in many of the areas most affected by civil conflict, the threat to agriculture posed by climate change, and the severe economic and human harms of civil conflict (see e.g., Blattman and Miguel, 2010; Fang et al., 2020). Much of this literature focuses on short-term impacts and is motivated by the idea that negative agricultural shocks increase conflict risk by lowering the opportunity cost of fighting.¹ This standard interpretation implies that negative shocks should directly affect conflict when they occur, as agricultural producers have less to lose by engaging in violence. Yet violent conflict is typically collective, organized, and shaped by broader social and political conditions. This paper analyzes whether severe transitory shocks to agricultural production have long-term effects on violent conflict, and whether any such effects operate by triggering immediate conflict onset or by dynamically amplifying later conflict incidence under broader instability.

Outbreaks of desert locust swarms provide a useful setting for answering these questions. Desert locusts are the world’s most dangerous and destructive migratory pest due to their potential to form massive cohesive swarms (Cressman et al., 2016; Lazar et al., 2016), which effectively constitute an agriculture-specific natural disaster. Climate change is creating conditions more conducive to swarm formation (Qiu, 2009), potentially undoing progress from increased international monitoring and control efforts in recent decades. The arrival of a locust swarm often leads to complete destruction of agricultural production and other vegetation (Symmons and Cressman, 2001; Thomson and Miers, 2002). Swarm flight patterns create quasi-random variation in the areas exposed to agricultural destruction in a swarm’s migratory path, and their migratory nature means that exposure to a swarm does not mechanically increase future risk from locusts. Because outbreaks in a given region have also been relatively infrequent over the last several decades, first observed swarm exposure provides a useful natural experiment for analyzing how rare, transitory agricultural production shocks affect the long-term risk of violent conflict.

Using data on the location and timing of desert locust swarm observations from the Food and Agriculture Organization of the United Nations (FAO) and conflict events from the Armed Conflict Location & Event Data Project (ACLED) and Uppsala Conflict Data Program (UCDP), I estimate the effects of swarm exposure on annual violent conflict risk in 0.25° (around 28×28 km) grid cells between 1997-2018 across Africa and the Arabian Peninsula.² I define event time relative to the

¹Several studies find that shocks to agricultural prices increase conflict incidence (e.g., Dube and Vargas, 2013; Fjelde, 2015; McGuirk and Burke, 2020; Ubilava et al., 2022). Impacts on agricultural productivity are speculated to explain the widely-studied relationship between climate or weather shocks and conflict risk (see Burke et al. (2015, 2024), Carleton et al. (2016), Dell et al. (2014), Hsiang and Burke (2013), and Koubi (2019) for reviews), though weather may affect conflict through mechanisms other than agriculture and some studies find results that are not consistent with effects through agricultural productivity (e.g., Bollfrass and Shaver, 2015; Sarsons, 2015) and many reviews point to a need for more evidence on mechanisms.

²I include all countries where at least 10 locust swarms are reported during the sample period. Tornrgren Wartin (2018) estimates short-term impacts of desert locusts on conflict in Africa using similar data. That paper focuses on potential measurement issues which I discuss in Sections 2.3 and 4.2, and does not consider long-term impacts of

first year in which a swarm is observed in a cell after 1989, and estimate how conflict risk changes in the years after that first exposure. This definition does not imply that swarm presence persists, as the actual swarms are transitory, but that their economic and conflict consequences may unfold over time after the swarms have gone. I estimate both average post-exposure effects and dynamic event study effects using the Borusyak et al. (2024) estimator to account for staggered treatment timing and potential heterogeneous effects, and test the sensitivity of the results to using other difference-in-differences estimators.

Locust swarm exposure increases the annual probability of any violent conflict event occurring in a 0.25° grid cell by 1.8 percentage points (64%) on average in years after exposure, compared to unaffected areas in locust migration paths in the same country. This average effect masks a central dynamic pattern: conflict risk does not increase in the year of exposure or the following year, when the direct negative shock to agricultural production should be strongest. Instead, effects emerge with delay and remain large up to 14 years after exposure. The absence of immediate effects is difficult to reconcile with a standalone short-run opportunity cost mechanism as a sufficient explanation for the impacts of this severe agricultural shock on conflict. At the same time, the effects are entirely driven by cells with crop or pasture land and by swarms arriving in crop cells during the main growing or harvest season, indicating that impacts operate through the initial agricultural destruction. I find limited evidence of conflict spillovers outside exposed cells, and the results are robust to a variety of alternative specifications.

The delayed effects are concentrated under broader instability. The largest effects on violent conflict risk occur 7-10 years after swarm exposure, aligning with the gap between the main locust exposure event in 2003-2005 and the emergence or escalation of conflicts in the sample countries associated with the Arab Spring, multiple civil wars, and the spread of terrorist organizations. Consistent with this timing, the impacts of past swarm exposure are concentrated in periods and places with greater surrounding civil conflict and insecurity. These results do not imply that locust shocks cause the later onset of broader conflicts. Rather, they indicate that prior exposure increases conflict incidence when broader instability, which may emerge for many reasons, makes violence more likely. Severe agricultural shocks therefore need not generate immediate conflict on average, but can amplify conflict risk when proximate conflict processes are active and either reduce the costs or increase the returns to fighting. This distinction matters because it separates the effects of shocks on conflict incidence from the origins of the broader conflicts in which those effects are realized.

To interpret these patterns, I develop a simple framework building on models of occupational choice between productive activities and armed conflict (Chassang and Padró i Miquel, 2009; Dal Bó and Dal Bó, 2011; McGuirk and Burke, 2020). I extend the basic framework in two ways. First, I allow a transitory agricultural shock to have persistent effects through a wealth or *permanent income* channel in which costly coping responses reduce human, physical, or financial capital, lowering future productivity and therefore the future opportunity cost of fighting. Second, building

locust swarms or mechanisms that are the main contributions of this paper.

on models emphasizing how social and political conditions shape conflict participation (Buhaug et al., 2021; Collier and Hoeffler, 2004), I allow the broader instability or mobilization environment to affect when lower opportunity costs translate into conflict participation. This motivates a *conditional-amplification* logic: past shock exposure may increase conflict incidence primarily when broader conflict processes are already active.

The empirical patterns discipline the interpretation of mechanisms in several ways. Null immediate effects are inconsistent with a simple short-run opportunity cost mechanism as a standalone explanation, while comparisons across conflict types provide little support for offsetting changes in the returns to predatory conflict as the explanation for the short-run null. The null short-term effects also make conflict hysteresis unlikely to explain the long-term increases in conflict. The long-term effects and agricultural specificity of the results are consistent with persistent opportunity cost effects, but additional tests provide only limited and indirect evidence of this channel. I find no significant long-term effects on grid-cell measures of the Normalized Difference Vegetation Index (NDVI) or remotely-sensed crop yield estimates, but do find significant decreases in survey-reported crop area cultivated and suggestive increases in out-migration. Together with evidence from other studies showing persistent adverse effects of locust swarms on agricultural production and human capital using survey data, these results make long-run opportunity cost effects plausible but not conclusively identified. Other persistent economic, social, or political effects of swarm exposure may also contribute to delayed conflict impacts.

To assess whether these dynamic and conditional patterns are specific to locust swarms, I analyze exposure to drought in the same econometric framework using monthly Standardized Precipitation and Evapotranspiration Index (SPEI) data. I find large time-varying increases in conflict risk following drought exposure that are also concentrated in locations experiencing conflict in surrounding areas. These results suggest that delayed and instability-contingent conflict responses are not unique to the biology, monitoring, or institutional context of desert locust swarms.

The long-term impacts also have econometric implications for empirical studies of economic shocks that may have persistent effects. This is particularly relevant for the economic shocks and conflict literature, where many studies define shocks as transitory and estimate only contemporaneous or short-term effects using two-way fixed effects. Such analyses are misspecified when shocks have non-zero average long-term effects, because later post-shock years are treated as comparison periods rather than as potentially affected periods. For both locust swarms and drought, specifications that impose transitory effects produce short-term estimates that are smaller than, and in some cases opposite in sign to, estimates from event study specifications that allow effects to evolve over time. This shows that short-term specifications can obscure the timing and persistence of shock effects, leading researchers to misinterpret both the mechanisms and policy implications of transitory shocks.

This paper makes three contributions. First, I contribute to the economics literature on conflict by providing evidence that severe transitory agricultural shocks can have long-term effects on conflict risk without directly triggering conflict when they occur. A large literature links cli-

mate, weather, and agricultural shocks to conflict risk and often emphasizes opportunity cost interpretations in which negative agricultural shocks increase conflict risk by reducing the returns to agricultural labor, though the mechanisms driving these relationships remain contested (see Burke et al. (2024) for a recent review). Using variation in exposure to desert locust swarms, I show that this interpretation is incomplete on its own. The absence of immediate effects does not align with a standard short-run opportunity cost prediction, while the concentration of later effects under broader instability points to a conditional pattern of conflict amplification rather than direct conflict onset. This builds on evidence that economic shocks can interact with social and political conditions or ongoing conflicts (Berman and Couttenier, 2015; Buhaug et al., 2021). I shift the empirical focus from contemporaneous conflict responses to how past shocks shape the geography and intensity of later violence when other conflict processes are active.

Second, I contribute to the broader literature on the long-term impacts of environmental shocks and natural disasters. Many papers show that adverse environmental shocks can have persistent effects on poverty and well-being (Baseler and Hennig, 2023; Carter and Barrett, 2006; Carter et al., 2007; Lybbert et al., 2004), but studies of agricultural disasters and conflict have focused primarily on short-term effects.³ I study conflict as a long-term outcome of disaster exposure and show that effects can be large, dynamic, and shaped by later social and political conditions rather than concentrated only in the short term. While the paper’s main identification strategy exploits quasi-random exposure to locust swarms, the results for drought exposure suggest that similar delayed and instability-contingent patterns may arise for other severe agricultural shocks. The results also show that empirical designs estimating only contemporaneous or short-term effects of severe economic shocks may not only miss important post-shock dynamics but also bias estimated short-term effects, distorting mechanism interpretation and policy implications.

Third, this paper adds a new dimension to studies of the economic impacts of agricultural pests (see e.g., Banerjee et al., 2010; Bradshaw et al., 2016), including desert locusts (see e.g., Thomson and Miers, 2002). The range of many agricultural pests is expanding due to climate change and globalization, and desert locusts are particularly likely to benefit from climate change (Qiu, 2009). A growing recent literature using survey data finds that locust swarm exposure adversely affects long-term education (De Vreyer et al., 2015; Yu et al., 2026), health (Conte et al., 2023; Gantois et al., 2024), and agricultural production (Marending and Tripodi, 2022) outcomes. I build on this evidence but study a different outcome: violent conflict, which depends not only on exposure to economic harm but also on the surrounding conflict environment. The results on long-term effects on conflict risk add a new dimension to estimates of the social costs of agricultural pests and should be considered when evaluating the benefits of prevention, monitoring, and post-shock recovery.

³One exception is Narciso and Severgnini (2023), who show that individuals in families more affected by the Great Irish Famine were more likely to participate in the Irish Revolution decades later. To my knowledge, no other study has explored long-term impacts on conflict risk of a transitory negative shock to agricultural production. Crost et al. (2018) and Harari and La Ferrara (2018) estimate effects of weather shocks on conflict over 2 and 5 years. Iyigun et al. (2017) consider long-run effects on conflict risk of a *positive and permanent* agricultural productivity shock from the introduction of the potato to the Eastern Hemisphere.

2 Setting and data

2.1 Desert locust swarms as a transitory agricultural shock

Desert locusts (*Schistocerca gregaria*) are a species of grasshopper always present in small numbers in desert ‘recession’ areas from Mauritania to India.⁴ They usually pose little threat to livelihoods, but favorable climate conditions in breeding areas—periods of repeated rainfall and vegetation growth overlapping with the breeding cycle—can lead to exponential population growth and the formation of large mobile swarms (Symmons and Cressman, 2001). Desert locust swarms vary in density and extent, but include billions of locusts covering tens of square kilometers on average; about half exceed 50 km² in size (Symmons and Cressman, 2001; FAO and WMO 2016). When swarms migrate away from breeding areas and affect multiple countries, this is referred to as an outbreak or upsurge. Although the frequency of major outbreaks has fallen in recent decades due to increased monitoring and control, climate change is expected to increase the risk of desert locust swarm formation and upsurges (Qiu, 2009; Youngblood et al., 2023).

Farmers have no proven effective recourse when faced with the arrival of a locust swarm (Dobson, 2001; Hardeweg, 2001; Thomson and Miers, 2002). The only current viable method of swarm control is direct spraying with pesticides, which is slow and costly to organize and requires robust locust control infrastructure (Cressman and Ferrand, 2021). Such control operations may limit swarm spread but generally cannot prevent immediate agricultural destruction where swarms arrive.

The characteristics of desert locust swarms make them a useful natural experiment for analyzing the long-term impacts of agricultural production shocks. First, the timing of upsurges and patterns of swarm flight create quasi-random temporal and local variation in swarm exposure. Locust swarms are migratory and fly 9-10 hours per day, generally downwind, easily moving 100 km or more per day even with minimal wind (FAO and WMO 2016). Conditional on being in the migratory path during an upsurge, some areas are exposed while others are flown over and spared any damages.⁵ The arrival of a swarm also does not change future risk (Figure D3 shows this empirically), so the direct shock to agricultural production is transitory even if indirect effects may persist.

Second, the arrival of a locust swarm is effectively a locally and temporally concentrated natural disaster affecting crops, pasture, and other vegetation (Hardeweg, 2001), but not directly affecting other aspects of the economy. This contrasts with temperature and precipitation shocks, which may affect infrastructure or physiology as well as agricultural production. The arrival of a swarm can lead to the total destruction of local vegetation (Symmons and Cressman, 2001; Thomson and Miers, 2002). The median swarm consumes 8 million kg of vegetation per day and swarms feed across many types of crops and pasture (FAO, 2023a; Lecoq, 2003). For example, during the 2003-2005 locust upsurge in North and West Africa, losses of 100, 90, and 85% were recorded for cereals, legumes, and pastures respectively, affecting more than 8 million people (Showler, 2019).

Third, the severity of agricultural destruction and limited tools for farmers to prevent damages

⁴Additional detail on desert locusts is included in Appendix C. Any time I use ‘locusts’ in this paper I am referring exclusively to desert locusts.

⁵Figure D5 illustrates the local and temporal variation in exposure to swarms for the area around Mali.

create potential for persistent adverse effects on well-being and conflict risk. Households exposed to locust swarms commonly report selling animals and other assets, consuming less food, sending household members away, taking loans and cutting expenses, and consuming seed stocks as coping strategies (Thomson and Miers, 2002). A swarm exposure shock therefore represents a shock to income and household wealth as well as agricultural productivity in the year of exposure.

2.2 Data

The Locust Watch database (FAO 2023) catalogs observations of desert locust swarms as well as smaller concentrations of locusts from 1985 to 2021. I consider only data on locust swarms, which pose the greatest threat to agriculture and whose flight patterns create local variation in exposure. The Locust Watch data include latitude, longitude, and date of swarm observations. Locust observations are recorded by national locust control and monitoring officers on the ground, but incorporate reports from agricultural extension agents, government officials, and other sources. Local farmer scouts are also often trained in locust monitoring and reporting (Thomson and Miers, 2002). Figure 1 displays the locations of all recorded desert locust swarm observations for the area of interest for this study.⁶ As illustrated by the figure, most observations are concentrated during a few periods of major upsurges.

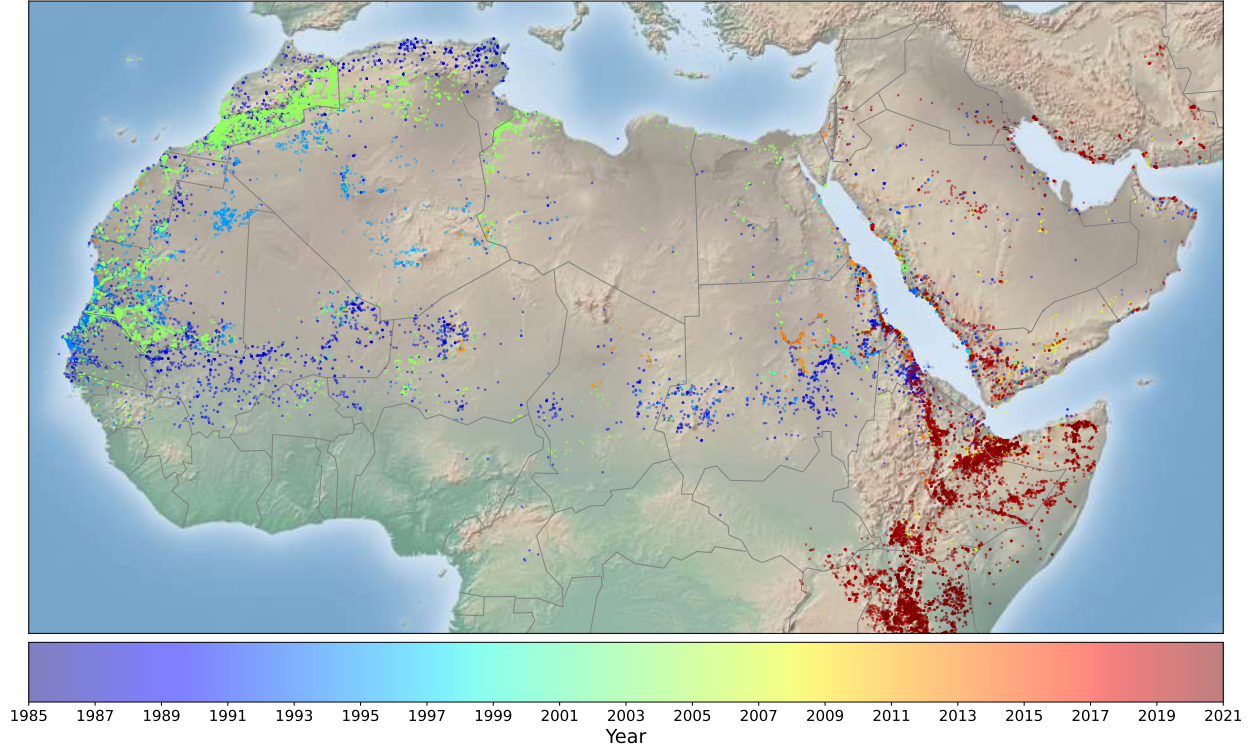
Data on conflict events come from the Armed Conflict Location & Event Data Project (ACLED) database (Raleigh et al., 2010). The database records the location, date, actors, and nature of conflict events globally starting from 1997 by compiling and validating reports from traditional media at different levels, from institutions and organizations, from local partners in each country, and from verified new media sources. The database includes many types of events including protests, riots, and various conflict-related strategic developments, but I focus on events categorized by ACLED as “violent conflict”: battles, explosions, and violence against civilians.⁷ As an alternative conflict type, I follow McGuirk and Burke (2020) in constructing a measure of output conflict (i.e., banditry) using ACLED records of violence against civilians, rioting, and looting. Such smaller-scale output conflict aligns well with the modeling of violent conflict as an occupational choice, and motivates using ACLED as the main source of conflict data.

For robustness, I also consider conflict data from the Uppsala Conflict Data Program (UCDP; Sundberg and Melander, 2013). The UCDP database uses similar sources as ACLED but goes back to 1989 and only records conflicts involving at least one “organized actor” and resulting in at least 25 battle-related deaths in a calendar year. The ACLED database has no organized actor or minimum death threshold requirements. The UCDP database has the advantage of being less subject to underreporting bias: some research has shown that events with no fatalities—including

⁶Desert locust swarms also affect other countries in the Middle East and South Asia, but not during the time period of this study.

⁷I provide examples of each from 2012 in Mali. Battle: On August 22 in Ansongo, MUJAO fighters and Military forces from Niger clashed close to the town of Tessit, with four MUJAO fighters killed. Explosion: On August 15 in Ansongo, anti-personnel mines planted by MUJAO exploded and killed two people. Violence against civilians: On July 26 in Timbuktu, a shepherd was killed after resisting an attempt by Ansar Dine fighters to steal one of his animals.

Figure 1: FAO Locust Watch swarm observations by year



Note: Map created by author using locust swarm records in the FAO Locust Watch database. Each dot represents the location of a single locust swarm observation, and the color indicates the year of the observation.

many of the types of ACLED violent conflict events I consider—suffer from more underreporting (Croicu and Eck, 2022; Eck, 2012). While UCDP captures a different type of violent conflict from ACLED, it therefore presents a useful check of the results.

I collapse these point event data to a raster grid with annual observations for cells with a 0.25° resolution (15 arcminutes, approximately 28×28 km). Analyzing impacts at this spatial level reduces potential measurement error about the specific areas affected by swarm and conflict events and allows me to leverage local variation in swarm presence created by their flight patterns. The median swarm covers around 50 km^2 , so nearly all swarms will be contained within 0.25° cells ($\sim 784 \text{ km}^2$), except those near cell boundaries. I test for robustness to analyzing data at the level of 0.5° and 1° cells, which will also capture potential spillovers from swarm exposure. In each cell and year I measure whether any locust swarm and conflict event is recorded. I do not account for variation in the counts of swarm or conflict events as the individual events are not themselves of consistent magnitudes. To test for spatial spillovers, I also measure whether any swarms are observed within 100 km outside of each cell-year.

I determine the country and highest sub-national administrative unit in which each cell centroid lies using country boundaries from the Global Administrative Areas (2021) database v3.6. I use sub-national administrative boundaries to create a set of 285 regions, all of which include at least 32 individual grid cells except for small countries with fewer than 32 cells. These regions are

either existing sub-national administrative units or combinations of adjacent units within the same country. I cluster standard errors at the level of these regions.

Given the role of weather in desert locust biology, its importance in determining agricultural production, and the well-documented relationship between weather shocks and conflict, all analyses control for local weather to isolate the impact of locust swarm exposure. I measure total annual precipitation (in mm) and maximum temperature (in °C) using high-resolution monthly data from WorldClim available through 2018.⁸ I measure drought exposure using monthly Standardized Precipitation and Evapotranspiration Index (SPEI) data from the Global Drought Monitor (Begueria et al., 2014). My primary definition of drought at the annual level is at least 4 consecutive months in a year where the SPEI in a cell is below -1.5, indicating significantly drier conditions relative to local historical norms (values from -1 to 1 indicate near-normal within-cell conditions).

I also incorporate raster population data for every 5 years from CIESIN, 2018, linearly interpolating within cells between years where the population is estimated, and raster data on land cover in 2000 from CIESIN, giving the share of land cover in each cell that is cropland and pasture (Ramankutty et al., 2010). I combine the land cover data with cropland mapping of Africa from 2003-2014 (Xiong et al., 2017) to identify cells with any cropland during the study period. I include additional cell characteristics, such as the start and end month of the main agricultural season, from the PRIO-GRID dataset (Tollefsen et al., 2012), assigning all 0.25° cells the values for the 0.5° PRIO-GRID cell in which they are located.

For the analysis of mechanisms, I incorporate data on agricultural production, economic activity, and net migration. Household-level estimates of agricultural production come from the Advancing Research on Nutrition and Agriculture (AReNA) Demographic and Health Surveys (DHS) GIS database (IFPRI 2020), which includes geolocated data at the level of household survey clusters for 40 surveys from 9 countries in the study sample conducted between 1992 and 2018. I incorporate two satellite-based measures of agricultural productivity: the Normalized Difference Vegetation Index (NDVI) and estimates of major crop yields. I calculate the NDVI—a commonly-used satellite-based measure of vegetation greenness at a given point and time—using 16-day 1km satellite imagery from MODIS (Didan, 2015) for the period 2000-2019, taking the maximum of monthly means in each grid cell to construct an annual value. In crop land, NDVI can be considered a rough proxy for agricultural production. Global annual yield data for four major crops—maize, rice, wheat, and soybean—at the 5 arcminute resolution for 1982-2015 come from Cao et al. (2025). Most cells only include data for one of the four crops, but for cells with multiple crops in the dataset I define the ‘main’ crop as the crop with the highest yield. Finally, net migration at the 5 arcminute resolution for 2000-2019 come from Niva et al. (2023), who estimate net migration based on subnational annual data on population, births, and deaths.

⁸CRU-TS 4.03 (Harris et al., 2014) downscaled with WorldClim 2.1 (Fick and Hijmans, 2017). I test sensitivity to measuring rainfall using CHIRPS (Funk et al., 2015) and temperature using ERA5 (Hersbach et al., 2019) to account for satellite-based weather measurement error (Josephson et al., 2024).

2.3 Locust swarm monitoring

The Locust Watch database likely does not include all locations of swarm exposure events over time due to monitoring capacity limitations. Randomly missing swarm events—classical measurement error—would attenuate estimated effects, but monitoring may also be correlated with characteristics related to conflict risk, such as agricultural activity, population, accessibility, state capacity, or insecurity. For example, Gantois et al. (2024) find heterogeneity in locust reporting across country borders, indicating differences in country monitoring capacities. Defining exposure as an annual dummy variable at the grid-cell level reduces this concern because only one swarm needs to be reported in a relatively large area to define the cell as exposed, but differences in cell-level monitoring effort may still lead to biased estimates.

The main empirical specification addresses this concern in several ways. First, I restrict the sample to cells where a locust swarm was ever reported within 100 km, dropping cells with no real risk of swarm exposure as well as cells far from any monitoring activity. Second, I control for population and weather variables likely correlated with both conflict risk and monitoring intensity. Cell fixed effects control for fixed characteristics which may affect monitoring and locust exposure risk, such as elevation, distance from breeding areas, and typical swarm migratory paths. Country-by-year fixed effects control for average differences in state monitoring capacity.

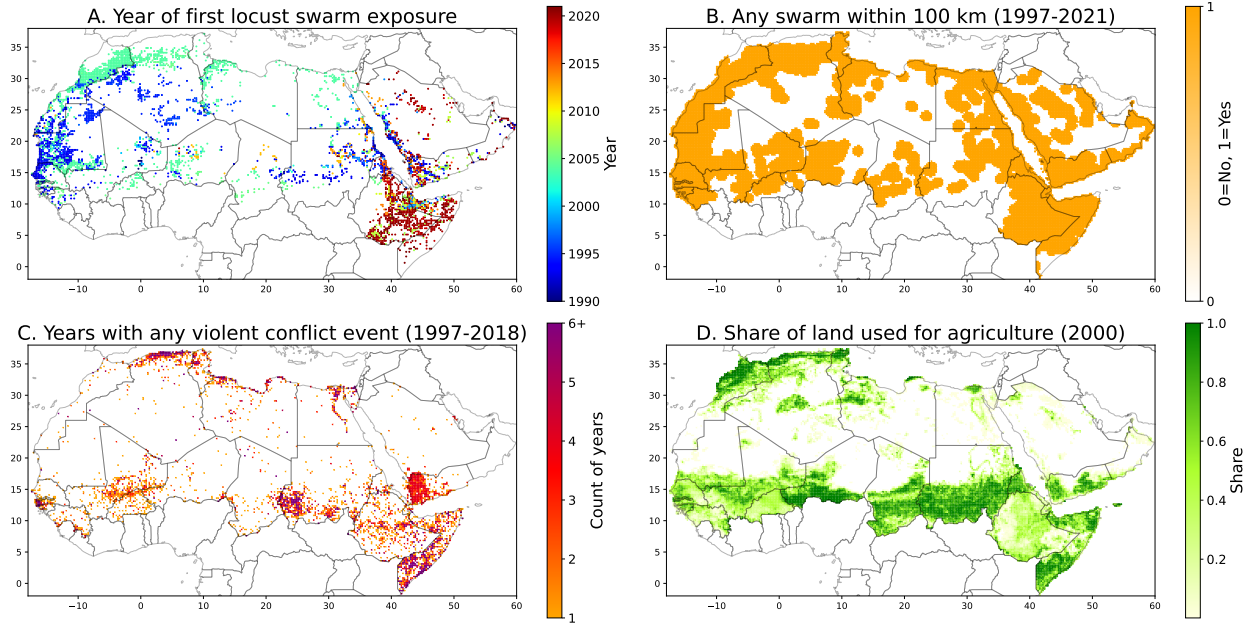
A specific concern is that insecurity may affect locust reporting. This concern motivates Torngren Wartin (2018)’s analysis of the impact of locusts on conflict, which uses similar data but focuses on the short term. Showler and Lecoq (2021) review how insecurity has affected desert locust control operations, though insecurity is likely less of a constraint for monitoring than for control. FAO monitoring guidelines discuss aerial surveys and reports from local scouts, agricultural extension agents, security forces, and other sources (Cressman, 2001), which would allow reporting even in insecure areas. Gantois et al. (2024) find that contemporaneous conflict reduces the probability of any locust monitoring, but generally has no significant effect on locust swarm reporting.

In Sections 4.2 and 4.3, I test whether the results are sensitive to excluding areas where insecurity may have affected swarm reporting, imputing potential missing swarms near reported swarm activity, aggregating to larger cells, controlling flexibly for estimated propensity of swarm exposure and other geographic characteristics, and systematically excluding different countries and regions from the sample.

2.4 Sample and summary statistics

Since ACLED records conflicts beginning in 1997 and the main weather data are available until 2018, the analysis sample includes observations from 1997 to 2018. I restrict the analysis to countries with at least 10 locust swarm observations in this period. The resulting analysis sample covers 22 years across 25,435 cells, for a total of 557,018 observations with data on all main estimation variables. Figure 2 visualizes swarm exposure, violent conflict incidence, and agricultural land cover for the sample countries. Summary stats are included in Table A1.

Figure 2: Swarm exposure, violent conflict incidence, and land cover in sample countries



Note: Panel A presents the first year after 1989 in which a locust swarm is observed in a given cell, using FAO Locust Watch data. This defines the swarm exposure year. Panel B shows all cells where any locust swarm was ever observed within 100 km of the cell centroid from 1996-2021. Panel C shows the count of years for each cell in which any violent conflict event was recorded in the ACLED database, from 1997-2018. Panel D presents 'baseline' land cover data for the year 2000 from CIESIN. Land used for agriculture includes crop land and pasture land. T

Locust swarms are relatively rare events, with swarms reported in less than one percent of cell-years (Table A1 Panel A). But at least one locust swarm is recorded in 9% of cells in the study period of 1997-2018 and 55% are within 100 km of any locust swarm report (Figure 2 Panel B). To account for the possibility of persistent effects of swarm exposure, I identify for each cell the first year after 1989 in which a locust swarm is recorded (Figure 2 Panel A), and define a cell as exposed to a locust swarm in each following year and not exposed in all other years or if no locust swarm is ever observed. Cells first exposed to a swarm before 1997 (in dark blue in Figure 2 Panel A) are considered treated during the entire sample period and therefore do not inform the analyses, while cells first exposed to a swarm after 2018 (in red) are considered not treated during the sample period. Just over seven percent of cells are first exposed to a swarm during the sample period, including 5.3% exposed during the 2003-2005 upsurge (in teal).

Violent conflict is also uncommon, with events reported in two percent of cell-years (Table A1 Panel A). Conflict is temporally correlated (Figure 2 Panel C), with 13% of cells experiencing at least one violent conflict event during the study period, in 3.4 different years on average (Table A1 Panel B). The risk of any violent conflict is fairly low from 1997-2010 before increasing significantly over the remainder of the sample period (Figure 4 Panel A). The increase corresponds with the timing of the Arab Spring movements, the spread of Islamic militant groups, and multiple civil wars and separatist movements in the sample countries.

Over half the cells (57%) in the sample have agricultural land: 56% have pasture land while 31% have crop land. Across all cells, the mean share of land allocated to agriculture is 23% (Figure 2

Panel D, [Table A1](#) Panel B), with 18% pasture land and 5% crop land. Given that locust swarms should affect outcomes through agricultural destruction, I test for heterogeneity in impacts by land cover.

3 Empirical approach

I use a difference-in-differences approach to estimate the causal impacts of locust swarm exposure on violent conflict. For each cell, treatment timing is the first year after 1989 in which a locust swarm is observed. A cell is coded as untreated before that year and treated in that year and all following years; cells with no observed swarm remain untreated throughout the sample. The treatment effect of interest is not the effect of being permanently treated by locusts. The shock itself is transitory: a swarm damages local vegetation when it arrives, but exposure to that swarm does not mechanically persist in later years. Consistent with this, [Figure D3](#) shows that first swarm exposure is not meaningfully associated with future exposure risk beyond the immediately following period. The empirical design is intended to analyze possible indirect long-term consequences of locust swarm exposure on conflict risk.⁹

I estimate both average post-exposure effects and event-study effects by years since first exposure using the Borusyak et al. (2024) imputation estimator (BJS). I also consider alternative estimators including Callaway and Sant’Anna (2021), Cengiz et al. (2019), De Chaisemartin and d’Haultfoeuille (2024), and Sun and Abraham (2021) and a standard two-way fixed effects (TWFE) specification, and find similar results. I present a standard TWFE estimation model to build intuition on the setup and the source of identifying variation in the analysis, as this is the same with the BJS estimator.¹⁰ The static average impact TWFE linear probability model takes the form:

$$Y_{ict} = \alpha + \beta Exposed_{ict} + \delta X_{ict} + \gamma_{ct} + \mu_i + \epsilon_{ict} \quad (1)$$

where i indexes cells, c countries, and t years. The outcome Y in the primary specifications is a dummy variable for observing any violent conflict event in the cell during a given year in the ACLED database. Analyzing conflict as a binary variable at an annual level reduces potential measurement error and is the main approach in the climate and conflict literature. I consider effects on other outcomes in tests of the impact mechanisms. *Exposed* is a dummy for whether a cell has already experienced its first observed swarm exposure. It equals 0 before the first exposure

⁹Repeated direct exposure is rare in the sample, and [Table B7](#) shows similar dynamic patterns of effects when exposure histories are defined using time since the most recent swarm rather than first exposure.

¹⁰The BJS estimator I use includes the same fixed effects and controls, so leverages the same identifying variation. The process of the estimation is different, however, proceeding in three general steps (Borusyak et al., 2024, p. 3255). First, the estimator fits $\hat{\alpha}$, $\hat{\mu}_i$, $\hat{\gamma}_{ct}$, and $\hat{\delta}$ using only untreated observations. Next, these fitted values are used to impute untreated potential outcomes for treated cells post-exposure. These imputed outcomes are subtracted from actual outcomes to estimate treatment effects. Then, the estimator takes an average of treatment effect, either on average for the static impact estimate or by period relative to treatment for the dynamic estimate. Because the time fixed effects are estimated using only data from untreated observations, the $\hat{\gamma}_{ct}$ are implicitly weighted toward periods with more untreated units, but otherwise the averages are calculated using equal weighting across treatment cohorts (by year of exposure). In this study the untreated groups are quite large, so the implicit weighting should be fairly uniform.

year and switches to 1 in that year and in all later years. The dynamic version of the analysis then estimates how conflict risk evolves relative to that first exposure year, using 10 leads and 10 lags. I choose 10 years to account for the fact that the analysis sample focuses on the period 1997-2018 and just under three-quarters of swarm exposure in this period takes place in 2003-2005.¹¹ I test the robustness to considering different numbers of lags and leads.

Cell fixed effects μ_i control for time-invariant cell characteristics (over the sample period) which might affect both the likelihood of swarm exposure and of experiencing violent conflict. Country-by-year fixed effects γ_{ct} flexibly control for country-level variation over time in the intensity of locust swarms and violent conflict. X_{ict} is a vector of time-varying controls at the cell level to account for locally-varying conditions which may affect both swarm exposure and the risk of conflict. The main specifications include annual cell total precipitation, maximum temperature, and population. Impacts of swarm exposure are therefore estimated using differences within cells in variation over time in violent conflict, between cells in the same country and with the same precipitation, temperature, and population but with different exposure to locust swarms. Standard errors (SEs) are clustered at the sub-national region level (285 clusters) to allow for correlation in the errors within nearby areas over time.¹²

The identification strategy relies on quasi-random variation in which areas in the migratory path of locust swarms during a given outbreak are actually exposed to locust destruction, due to swarm flight patterns. The key identifying assumption of the econometric design is that trends in conflict risk would be parallel over time in exposed and unexposed areas within the same country in the absence of locust swarm exposure, after controlling for effects of weather and population.

Because the paper studies conflict dynamics over a long horizon, this identifying assumption is more demanding than in designs focused only on contemporaneous or short-run responses. I therefore place particular weight on pre-exposure dynamics, agricultural placebo comparisons, alternative conflict data with a longer pre-period, treatment cohort and country-group exclusions, and robustness to flexible geographic trends, and present results of these analyses following the main results.

The key threat to the assumption is if locust swarms are more likely to affect areas with different propensity to experience conflict. The literature on locust biology suggests that—conditional on swarms forming in breeding areas and then beginning to migrate—where swarms land is driven by wind direction and time of day rather than land cover or particular types of vegetation. This implies quasi-randomness in exposure within swarm migratory paths. But selection in locust reporting, discussed in [subsection 2.3](#), could lead to a violation of the parallel trends assumption by leading to differences in where swarms are observed.

While the parallel trends assumption is not possible to test directly, I explore its plausibility in

¹¹Because this concentration could make event time difficult to separate from calendar-time conflict dynamics, I report robustness checks below that drop the 2003–2005 exposure cohorts and that exclude groups of countries experiencing major subsequent conflict increases.

¹²Patterns of statistical significance are largely unchanged when using two-way clustered errors, country clustered errors, or Conley (1999) Heteroskedasticity and Autocorrelation-Consistent (HAC) SEs ([Figure B4](#)).

two main ways: testing for balance in baseline or fixed cell characteristics and testing for parallel trends in outcomes prior to exposure. I discuss balance in cell characteristics here and discuss pre-trends when presenting the main event study results. Cells exposed to a locust swarm during the sample period have different baseline characteristics than unexposed cells which are largely consistent with desert locusts rarely being observed in the interior of the Sahara desert (as shown in Figure 2). Exposed cells have larger populations, are closer to capital cities, have a greater share of pasture land and smaller share of barren land, and have lower maximum temperatures (Table A2). These differences are smaller and lose some statistical significance when restricting the sample to cells within 100 km of any locust swarm from 1997-2021, a proxy for being in potential swarm migratory paths (a joint test of differences in cell characteristics yields $F = 2.38$ and $p < 0.01$). These patterns appear more consistent with selection in locust swarm reporting than vegetation-driven selection in where swarms land, particularly as there is no difference in the share of cell crop land by exposure status.

One approach to account for imbalance in covariates is to weight observations by the conditional probability of being treated (Abadie, 2005; Baker et al., 2025; Stuart et al., 2014). I estimate the propensity of any locust swarm exposure during the study period as a function of fixed cell characteristics including land cover and population in 2000, distances to the capital and to a national boundary, mean weather realizations over the study period, and country fixed effects. I first use a lasso regression to select parameters and then estimate exposure propensity using a logit regression. I use the results to construct inverse propensity weights as $\frac{1}{p}$ for cells that were exposed and $\frac{1}{1-p}$ for cells that were not, where p is the estimated probability of swarm exposure. I assign cells with estimated probabilities outside the range of common support a weight of 0. Restricting to cells in likely migratory paths and including these weights largely eliminates differences in baseline characteristics ($F = 0.64$ and $p = 0.824$ for the joint test of differences; Table A2). The Borusyak and Hull (2023) approach to dealing with non-random exposure to exogenous shocks involves adjusting for average treatment across shock counterfactuals—this has been used in the case of locust swarms by Marending and Tripodi (2022). I draw on this by testing the robustness of the results to controlling for swarm exposure propensity-by-year fixed effects, which would account for potential different trends in areas with high exposure propensities, treating this as a proxy for average treatment under different shock realizations.

Baseline differences by swarm exposure status are not a concern if they do not affect conflict risk or only affect levels of conflict, but it is plausible that some of the differences would affect conflict trends. However, the controls in the empirical specifications should absorb most of these differences. Cell fixed effects control for time invariant cell characteristics that might affect the risk of conflict such as distance to major cities or country boundaries, topography, and agricultural suitability, but also distance from locust breeding areas. Country-by-year fixed effects flexibly control for factors varying over time at the country level that might affect conflict risk, such as food price shocks, weather patterns, the policy environment and national economic and social conditions. Importantly, they control for trends in violent conflict incidence, which increases over

the sample period. I also directly include in the regressions time-varying characteristics that differ between exposed and unexposed cells and may affect locust reporting and conflict risk—population and temperature—as well as annual precipitation. I control for baseline differences in distance to the capital by defining all cells as either below or above the median distance to the capital within each country, and including capital distance by year fixed effects in the regressions to account for potential differential trends in conflict risk over time by proximity to country capitals. Finally, the main analyses restrict the sample to cells within 100 km of any locust swarm from 1997-2021 (illustrated in [Figure 2](#) Panel B) and within the range of common support of the estimated propensity of swarm exposure across exposed and unexposed cells. This effectively restricts the sample to areas within swarm migratory paths during recent outbreaks, where exposure is most likely to be quasi-random.

Another identification assumption relevant to event study designs with staggered treatment timing is the no anticipation assumption: knowledge of future treatment timing does not affect current outcomes (Roth et al., [2023](#)). Populations may expect a higher probability of swarm exposure in years of major upsurges but cannot perfectly anticipate timing of exposure. For example, the FAO Desert Locust Watch publishes monthly forecasts of areas predicted to be at risk of locust swarm exposure but the predictions include a great deal of uncertainty due to unpredictable minor variations in swarm flight patterns. Consequently, areas forecast to be at risk are generally quite large, the majority of which end up not being affected by locusts.¹³ Anticipation may also have limited effects as there are no effective methods of defending vegetation against locust swarms, and farmers in at-risk areas typically describe locust prevention and control as out of their hands and the responsibility of governments (Thomson and Miers, [2002](#)).

Finally, the empirical specification assumes no effects of locust swarm exposure outside of the exposed cells. In the robustness checks, I first test the sensitivity of the results to conducting the analysis in larger grid cells, to increase the likelihood that any effects are contained within exposed cells. I then relax this assumption by including a spillover exposure indicator for being within 100 km of a locust swarm during the study period, defined similarly as the within-cell exposure variable.

4 Main results and identification tests

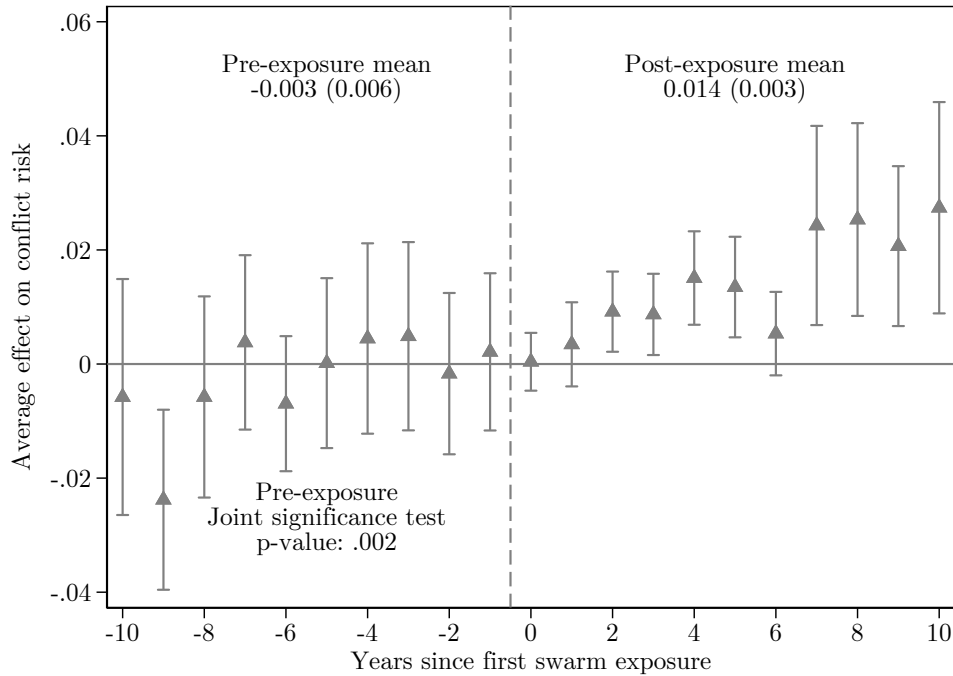
[Figure 3](#) presents event study estimates of the impact of desert locust swarm exposure on violent conflict at the cell-year level using the Borusyak et al. ([2024](#)) imputation estimator (BJS). The average pre-exposure difference is -0.003 but is not significantly different from 0, with five positive coefficients and five negative coefficients. Only one pre-exposure coefficient is larger than 0.007 in magnitude or statistically significant: on average violent conflict is 2.4 percentage points (pp) less likely in areas exposed to locust swarms compared to unexposed areas nine years before exposure. Because of this one highly significant difference I reject that pre-exposure differences are jointly

¹³I find that monthly forecasts of at-risk areas during the major upsurge in 2004 covered on average 40.6% of 0.25° cells in sample countries, but nearly one-quarter of swarms in this period were recorded outside of these areas.

equal to 0 ($p = 0.002$), but I fail to reject that the nine other pre-exposure coefficients are jointly equal to 0 ($p = 0.230$).¹⁴

As parallel trends is a key assumption for the identification strategy, I conduct additional tests of pre-trends. As the sample period spans 1997-2018 and the majority of swarm exposure occurs in 2004, I have less power to identify pre-treatment coefficients more than seven years before exposure. An event study with six pre-exposure periods shows similar pre-trend patterns but with tighter standard errors, and I fail to reject that the pre-exposure coefficients are jointly equal to 0 (Figure B1 panel A). I also consider results with a larger number of pre-exposure periods using UCDP data, which record conflict events starting in 1989. I find consistently negative but non-significant estimated effects over 15 years prior to exposure, and again fail to reject that pre-exposure coefficients are jointly 0 (Figure B2). I return to a comparison of results using ACLED and UCDP conflict data in subsection 6.2. Together, these patterns are consistent with similar pre-exposure conflict trends, though they cannot rule out the possibility that exposed and unexposed cells would have diverged later for reasons unrelated to swarm exposure.

Figure 3: Impacts of exposure to locust swarms on violent conflict risk over time



Note: The dependent variable is a dummy for any violent conflict event in a cell-year. Estimated impacts in each time period are weighted averages across effects for swarm exposure in particular years, calculated using the BJS estimator. Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. Observations are grid cells approximately 28×28km by year. The sample excludes cells more than 100 km from any locust swarm observation from 1997-2021 and outside the range of common support of the estimated propensity of swarm exposure across exposed and unexposed cells. Coefficient estimates are shown in Table B1 column 1. At the top left and right I show averages of pre- and post-exposure coefficients. On the bottom left I show the results of a joint test that the pre-exposure coefficients equal 0. A joint test omitting the period 9 years before exposure yields $p = 0.230$.

¹⁴A fitted line through the pre-exposure coefficients has a slightly positive slope, though smaller than the slope of the line through the post-exposure coefficients, but the pre-exposure line is close to flat if the -9 period is excluded.

In contrast with the pre-exposure coefficients, 8 of 11 treatment coefficients are statistically significant at a 95% confidence level. The average effect is a 1.4pp increase in the annual risk of any violent conflict event, but the effects are not consistent over time. There is no general immediate increase in violent conflict following swarm exposure. The point estimate in the year of exposure is a fairly precise 0, and the estimate for the following year is positive but not statistically significant. This challenges a central empirical prediction in the literature on agricultural shocks and conflict, where reductions in the opportunity cost of fighting are generally expected to raise conflict risk when the shock occurs, though Crost et al. (2018) and Harari and La Ferrara (2018) also find delayed effects for some growing season weather shocks. Instead, conflict responses emerge with delay. Estimated treatment effects become significant after the first post-exposure period and are largest 7-10 years after exposure, when all four coefficients are larger than 0.021 and significant at a 99% confidence level. The next section examines why we observe delayed effects on conflict risk.

Turning to average long-term impacts across all exposure and post-exposure periods, Table 1 presents estimated effects across all cells and by presence of agricultural land. On average cells exposed to locust swarms are 1.8pp more likely to experience any violent conflict in a given year in the period after swarm exposure than cells not exposed. This represents a 64% increase over the mean for non-exposed cells after 2004, the main year of swarm exposure, and corresponds to roughly 536 additional cell-years with violent conflict among exposed cells in post-exposure periods. The larger average long-term impact in Table 1 column 1 than in Figure 3 indicates that large magnitude effects on violent conflict persist more than 10 years after swarm exposure, and I show this when including 14 years of post-exposure effects in Figure B1 Panel B.

I find similar average long-term effects of swarm exposure estimated using TWFE as presented in Equation 1 (Table B3). The average long-term impact with this estimator is a 2.0pp increase, slightly larger than the main BJS estimate. This suggests limited bias in the TWFE estimates from staggered timing of swarm exposure, potentially because close to three-quarters of exposure occurred in the same period in 2003-2005.

The average long-term impact of swarm exposure is large compared to the same-year effects of weather fluctuations. A 1 SD increase in annual precipitation increases the probability of any violent conflict in the same year by 0.4pp (14%) compared to 1.2pp (43%) for a 1 SD increase in the maximum annual temperature. The effect of temperature is in the upper end of the distribution of estimates of the impacts of climate on intergroup conflict in Burke et al. (2024)’s meta-analysis, potentially because of the use of maximum temperature and the time period studied. Cell population is also positively associated with conflict risk, with an increase of 10,000 people associated with a 1.0pp (36%) increase in the probability of any violent conflict event in a year.

4.1 Placebo tests: Agriculturally relevant exposure

If estimated effects are capturing causal effects of swarm exposure we should expect heterogeneity by agricultural land cover and by the timing of exposure relative to crop production cycles. If on the other hand the analysis simply captures later conflict expansion in locust migratory paths, we should

Table 1: Average impacts of exposure to locust swarms on violent conflict risk by land cover

Outcome: Any violent conflict event	(1)	(2) Land = Any crop or pasture land	(3) Land = Any crop land
Average effect of swarm exposure	0.018*** (0.004)		
Total annual precipitation (SDs)	0.004** (0.002)	0.004** (0.002)	0.004** (0.002)
Max annual temperature (SDs)	0.012** (0.005)	0.012** (0.005)	0.012** (0.005)
Population (10,000s)	0.010*** (0.002)	0.010*** (0.002)	0.010*** (0.002)
Effect of exposure, Land=0		0.006 (0.006)	0.007 (0.004)
Effect of exposure, Land=1		0.019*** (0.005)	0.023*** (0.006)
Observations	301664	301664	301664
<i>p</i> -value, equality of effect by land cover		.109	.025
Outcome mean post-2004, no exposure	0.028	0.028	0.028
Country-Year FE	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Note: The table presents results from three separate regressions of average long-term impacts of locust swarm exposure on the probability of any violent conflict event in a cell-year using the BJS estimator. In columns 2 and 3 I estimate heterogeneous effects by land cover in 2000, considering whether there is any crop or pasture land and any crop land, respectively. The ‘Land=0’ row shows the estimated average long-term effects in cells with no such land cover while the ‘Land=1’ row shows the impact in cells with such land cover. At the bottom of the table I include *p*-values for the test of equality of these two coefficients. The outcome mean for control cells is shown for post-2004 for comparison with exposure impacts in the period after the majority of swarm exposure occurred. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

find no such heterogeneity. This suggests the possibility of a placebo test to evaluate whether the analysis is identifying a causal effect of swarm exposure and agricultural destruction, or a spurious effect of factors correlated with both swarm exposure reports and later violent conflict. Though it is not a pure placebo because land cover is measured imperfectly and could have changed over time, a particularly sharp falsification test is exposure to locust swarms in cells with no measured agricultural land during the off season or planting period. We should expect no effect in such cells, and significantly larger effects in cells with cropland where swarm exposure occurred during the main crop growing or harvest season.

I first test simply for heterogeneity by cell land cover (Table 1 columns 2 and 3). Swarm exposure in non-agricultural cells (23% of the analysis sample) has no significant effect while in agricultural cells annual violent conflict risk increases by 1.9pp, though I cannot clearly reject that the effects are the same ($p = 0.109$). Locust swarms increase annual violent conflict risk by 2.3pp in crop cells (57% of the sample) compared to no significant effect in non-crop cells (46% of which have pasture land), and here the difference is statistically significant ($p = 0.025$). These differences are consistent with locust swarms affecting conflict risk through initial agricultural destruction.

I find slightly larger differences in average long-term effects of swarm exposure by land cover using TWFE (Table B3), and the difference by any agricultural land becomes statistically significant at a 95% confidence level. The TWFE regressions also allow me to test heterogeneity in effects

of precipitation and temperature deviations (Table B3 columns 2 and 3), which is not possible with the Stata package implementing the BJS estimator.¹⁵ Effects of precipitation are marginally significantly larger in cells with any crop land but remain significant in non-agricultural cells. Effects of temperature do not vary by land cover. These results echo previous work questioning whether agricultural mechanisms alone can explain the relationship between weather and conflict (Bollfrass and Shaver, 2015; Sarsons, 2015).

To further test that impacts of locust swarms are driven by initial impacts on agricultural production, I consider heterogeneity in impacts by both land cover and by timing of swarm exposure relative to local crop calendars. I categorize swarms as arriving during particular stages of the crop production cycle by matching the month in which a swarm is observed to crop calendar information from the PRIO-GRID dataset, filling in missing data with country-level crop calendars from The United States Department of Agriculture (USDA) (2022). The off season—between harvesting and planting—lasts between 3 and 6 months in most of the sample cells, with an average of slightly over 4 months. I distinguish between swarms arriving during the off season and planting season (first two months of the main agricultural season) when they are unlikely to significantly damage crop production, and swarms arriving in the growing and harvesting season when potential damages should be greatest. Figure D4 presents the timing of locust swarms by region across the sample period.¹⁶ As with the main analysis, I use the first year a cell was exposed to a locust swarm during a particular season to define treatment timing. I then estimate Equation 1 with the two seasonal swarm exposure treatments using TWFE and fully interact all variables with dummies for any agricultural or crop land cover.

Table 2 column 1 shows that effects on violent conflict risk are driven by exposure to locust swarms during the growing or harvest season in cells with any agricultural land, where exposure increases average annual risk of any violent conflict event by 2.6pp. There are no significant effects of swarm exposure in other seasons or non-agricultural cells. The patterns are similar when considering differences by any crop land in particular (column 2). These results align with other studies showing that the impact of weather shocks on conflict risk varies depending on whether the timing of the shock is such that it is likely to decrease agricultural productivity (Caruso et al., 2016; Crost et al., 2018; Harari and La Ferrara, 2018).

Effects of placebo-like exposure - swarms in non-agricultural cells during the off season or planting - are close to 0. I cannot reject equality of effects by land cover of growing/harvest season exposure ($p = 0.161$ and $p = 0.127$, respectively for the two land cover dummies), but point estimates of the effect of growing or harvest season swarms in non-agricultural and non-crop cells are not statistically significant and are much smaller in magnitude. The positive point estimate in column 1 may reflect expansion of agricultural activity after 2000 (Xiong et al., 2017), while in

¹⁵The association between population and conflict risk does not vary significantly with land cover. The estimated effect in non-agricultural cells is very large but there is little identifying variation driving this estimate.

¹⁶Swarms in cells with crop land are more frequently observed during the growing/harvest season, likely because locust breeding primarily occurs during wetter months which tend to overlap with the planting period and swarm migration out of breeding areas follows in subsequent months.

Table 2: Impacts of swarm exposure by land cover and swarm timing

Outcome: Any violent conflict event	(1) Land = Any crop or pasture land	(2) Land = Any crop land
Off/planting season, Land=0	-0.001 (0.005)	0.001 (0.006)
Off/planting season, Land=1	0.007 (0.006)	0.009 (0.007)
Grow/harvest season, Land=0	0.007 (0.011)	0.012 (0.007)
Grow/harvest season, Land=1	0.026*** (0.007)	0.028*** (0.008)
Observations	304612	304612
Outcome mean post-2004, no exposure	0.028	0.028
p , off/plant season non-crop=crop effect	0.329	0.292
p , grow/harvest season non-crop=crop effect	0.165	0.127
p , non-crop off/plant=grow/harvest effect	0.541	0.257
p , crop off/plant=grow/harvest effect	0.059	0.089
Country-Year FE	Yes	Yes
Cell FE	Yes	Yes
Controls	Yes	Yes

Note: The table presents results from a single regression interacting two seasonal swarm exposure treatment variables with a dummy for crop land cover with the same fixed effects and controls as in [Equation 1](#). The coefficients and standard errors are calculated using Stata's *xtlcom* command based on the sums of the coefficients for the non-crop seasonal effects and the crop interaction terms. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

column 2 it may reflect effects through destruction of pasture or other vegetation, even if the main impacts are through crop destruction.

Crucially, for both land cover types (any agricultural land and any crop land specifically) I can reject equality of effects of swarm exposure in cells with such land cover by whether the exposure occurred in the off/planting seasons compared to the growing/harvest seasons ($p = 0.059$ and $p = 0.089$, respectively).¹⁷ We would not expect such heterogeneity unless impacts were truly driven by swarm exposure and crop destruction as opposed to characteristics of cells where locust swarms are reported.

4.2 Evaluating potential swarm observation bias

The heterogeneity in effects of swarm exposure by timing and land cover in [Table 2](#) increases confidence that the estimated impacts represent true causal impacts of swarm exposure and are not driven by potential bias in where locust swarms are reported or other spurious correlations. But measurement error in locust swarm exposure may still create bias. Other studies have raised concerns that swarms may be particularly underreported in insecure areas (Gantois et al., 2024; Showler and Lecoq, 2021; Tornngren Wartin, 2018). More generally, underreporting may be non-random if monitoring intensity differs across areas with different swarm risk, agricultural activity, population, accessibility, or conflict risk. The validation exercises in this subsection therefore evaluate whether the results are robust to several forms of missing-swarm measurement error.

¹⁷Small positive point estimates for effects of off/planting season swarms in such cells could reflect effects through destruction of vegetation even outside of peak crop production periods, such as perennial crops, early growth of annual crops, and pasture.

I first consider whether the results are sensitive to excluding areas where insecurity may have affected swarm reporting. Missing swarm observations in high-conflict areas should attenuate my estimated effects by misclassifying some conflict-prone areas as unexposed. Instead, average long-term effects are slightly smaller in magnitude when dropping countries where Showler and Lecoq (2021) indicate insecurity has limited locust control operations, and when dropping cells that experienced violent conflict during the 2003-2005 locust upsurge (Figure B7 Panel A). This suggests that effects of swarm exposure on conflict are larger in more insecure areas, a point I return to in the next section.

I then directly test the potential for conflict-related underreporting by imputing ‘missing’ swarms in a random share of cell-years with any locust swarm report within 100 km and any violent conflict within the cell. Estimated effects increase as I impute more ‘missing’ swarms in such areas (Figure C2 Panel A), consistent with attenuation from conflict-related underreporting. This does not imply that swarms are in fact substantially underreported in areas experiencing conflict, since larger estimates could also arise mechanically from assigning exposure to more conflict-prone cells.

I next test whether results are sensitive to missing swarm observations in cells with high swarm exposure propensity. For cells with a high estimated propensity of ever being exposed, I impute hypothetical ‘missing’ swarms in any year that an actual locust swarm is observed within 100 km. The event study results are very similar even if I assign all cells with an estimated propensity above 0.25 to have been exposed in the first year a swarm is reported nearby (Figure C1).

Finally, I randomly impute hypothetical ‘missing’ swarms across all cell-years with swarms reported within 100 km. This provides a broader bounding check for missing-swarm measurement error around observed swarm activity, without conditioning on estimated exposure propensity or conflict incidence. Estimated average long-term effects fall as I impute swarms in a larger share of nearby cell-years (Figure C2 Panel B), consistent with potential non-random reporting but also attenuation from added noise in the treatment definition. Even if I randomly assign 20% of cell-years near a swarm report to be exposed, the estimated effect remains economically and statistically significant, with a mean 0.6pp increase in violent conflict risk that is significant at the 95% level in 69% of simulations (Figure C2 Panel C). Taken together, these results provide strong evidence that the estimated effects of locust swarm exposure are unlikely to be driven entirely by selection in where swarms are reported.

4.3 Design, inference, and exposure robustness

I next report robustness checks targeting five aspects of the design: estimator choice and inference, flexible controls for geographic trends, dominance of the 2003-2005 exposure cohort, spatial scale and spillovers, and the definition of exposure timing.

I estimate event study effects using several alternative staggered difference-in-differences estimators (Callaway and Sant’Anna, 2021; Cengiz et al., 2019; De Chaisemartin and d’Haultfoeuille, 2024; Sun and Abraham, 2021) and find nearly identical dynamic treatment effects (Figure B3) and average long-term effects (Table B2). There is some variation in estimated pre-exposure differences,

which is expected as a key difference across these estimators is how pre-exposure effects are calculated. The average long-term effect is also robust to more conservative inference including two-way clustered errors at the year and sub-national region level, clustering at the country level, and using Conley (1999) Heteroskedasticity and Autocorrelation-Consistent (HAC) SEs allowing for spatial correlation over 100 and 500 km and serial correlation over 0 or 10 time periods, following Hsiang (2010)’s approach (Figure B4, Table B4).¹⁸

Event study results and average long-term impacts are similar when varying the fixed effects and controls included in the estimation (Figure B5, Figure B6). Estimated average effects are somewhat smaller (but not significantly different) when replacing country-by-year fixed effects with year fixed effects, and are if anything slightly larger when instead using sub-national region-by-year fixed effects. These more restrictive fixed effects identify effects of swarm exposure off of more local spatial variation, limiting concerns that results are due to comparing more to less conflict-prone areas. Variation in swarm exposure even within sub-national regions leads to large increases in violent conflict.

Effects remain significant and similar in magnitude when dropping the population, weather, and distance to capital-by-year controls, adding weather lags, using alternative measures of precipitation and temperature from CHIRPS (Funk et al., 2015) and ERA5 (Hersbach et al., 2019). The estimated magnitude of the average effects is only slightly smaller when adding a series of flexible geographic trends, including agricultural land-by-year, distance to nearest international border-by-year, baseline population-by-year, and estimated swarm exposure propensity-by-year controls. These controls help directly address potential concerns about selection in swarm reporting and conflict trends, with the swarm propensity-by-year controls in particular capturing a variety of factors which might make some cells more likely to have a reported locust swarm. In no case can I reject that the alternative estimate equals the main estimate.

I next consider sensitivity of the results to changing the analysis sample. The main specification excludes cells more than 100 km from any swarm report and outside the range of common support of estimated swarm exposure propensity, to focus on swarm migration corridors. The results are similar when including the full sample of cells. A separate concern is that the dynamic pattern of effects could be driven by the concentration of swarm exposure during the 2003–2005 upsurge and the subsequent rise in conflict in particular regions. I therefore test whether average effects are sensitive to treatment cohort and country-group exclusions. The average effect remains statistically significant and if anything is slightly larger when dropping cells first exposed in 2003–2005, and is similar when restricting treated cells to those first exposed in 2003–2005 while retaining never-treated comparison cells (Figure B7 Panel A). The dynamic pattern of swarm exposure impacts is noisier but broadly similar when the 2003–2005 cohorts are excluded (Figure B8).¹⁹ The main difference is that effects in years 4 and 5 post-exposure are now as large as the later effects, and

¹⁸Clustering at the sub-national region level consistently leads to SEs at least as large as Conley SEs allowing for spatial correlation within 500 km and serial correlation over 10 years, implying the main SEs I report are conservative and may understate statistical significance of certain estimates.

¹⁹Figure 4 Panel B shows an event study of exposure to the 2003–2005 upsurge specifically.

most of the later effects lose statistical significance at conventional levels. The estimates also remain positive and significant when excluding different regions and when excluding specific groups of countries affected by similar types of violent conflict over the study period: Egypt/Libya/Tunisia, Mali/Burkina Faso/Niger/Chad, and Somalia/Yemen/Sudan (Figure B7 Panel B). These tests indicate that the results are not mechanically driven by a single exposure cohort or by subsequent conflict increases in any particular country group.

I also consider differences in estimated effects when aggregating the sample to larger grid cells. Using larger grid cells addresses several potential measurement issues. First, it minimizes the possibility that the area exposed to a locust swarm recorded in a cell exceeds the boundaries of the cell. Second, it reduces concerns about nearby areas that might have been affected by unreported swarms since the entire cell is considered exposed if any swarm is reported within it. Third, it limits the potential for conflict spillovers outside the cell. Downsides to analyzing impacts in a more coarse grid are dilution of treatment intensity (as the share of the cell affected by swarms weakly decreases with cell size) and a reduction in the quasi-random local variation in swarm exposure which is central to the identification approach. The latter weakness implies that results at more aggregated cell sizes should be interpreted with caution. I observe similar patterns in dynamic impacts of swarm exposure over time when collapsing the data to 0.5° or 1° cells (Figure B9). Impacts in post-exposure periods follow a similar pattern as in Figure 3 but are larger in magnitude. Pre-exposure coefficients are more positive and generally larger in magnitude and some are marginally statistically significant. Looking at average long-term impacts, absolute effects are larger when collapsing the data to the level of 0.5 or 1° cells, but effects relative to the probability of any violent conflict in unexposed cells are larger in 0.25° cells (Table B5). This is consistent with higher probabilities of any violent conflict as cell size increases, and with less dilution of the swarm exposure treatment in smaller cells.

In addition to conducting the analysis at higher levels of spatial aggregation, I directly test for spillover effects of swarm exposure in nearby cells. Local agricultural destruction due to desert locust swarms implies both a sharp decrease in the opportunity cost of fighting for agricultural producers in affected areas and potential spillovers to nearby unaffected areas where the returns to fighting over output would be larger. The main analysis considers $\sim 28 \times 28$ km grid cells where the median locust swarm would only affect around 6% of cell area. Conflict incited by locust destruction may be more likely to be realized within exposed grid cells but large absolute effects of swarm exposure in larger grid cells (Table B5) suggests some conflict spillovers may yet occur. I define spillover exposure treatment as being within 100 km of a swarm outside of the cell and estimate effects of this treatment alongside direct exposure using the TWFE estimator. I find a fairly precise null average effect of this spillover treatment when controlling for direct swarm exposure, though spillovers from the 2003-2005 upsurge are marginally significant and indicate a 1.1pp average long-term increase in conflict risk for cells within 100 km of a swarm during that upsurge (Table B6). Focusing on impacts within 0.25° cells may therefore understate the full effect of swarm exposure on violent conflict risk around affected areas, but the results indicate that most

effects are contained within cells. This may relate to the fact that the main impacts of swarm exposure on conflict are delayed and therefore do not represent immediate responses of affected populations as in McGuirk and Nunn (2025) who find that drought in pastoral areas leads to conflict spillovers in nearby agricultural areas.

Finally, I test whether the delayed pattern of effects depends on defining treatment timing by *first* observed swarm exposure. I estimate TWFE specifications using annual direct swarm exposure histories, defining treatment bins by whether a cell was exposed in the previous 0–2, 3–6, 7–10, or 11 or more years. I obtain similar patterns when these windows are allowed to overlap (172 of 1833 exposed cells are exposed in more than one year in the study period) and when they are defined mutually exclusively based on time since the *most recent* swarm exposure (Table B7). Coefficients are small in the first two years after exposure and larger in later windows, indicating that the delayed profile is not an artifact of anchoring event time to first exposure.

5 Conditional conflict amplification under broader instability

The pattern of dynamic long-term impacts of locust swarm exposure on violent conflict shown in Figure 3 suggests that severe agricultural shocks need not independently cause the onset of violent conflict on average when they occur. But null immediate effects and larger effects after a delay of seven years are not an intuitive pattern, and importantly are not consistent with a standalone opportunity cost mechanism. A broader framework is therefore needed to explain when and why past swarm exposure becomes conflict-relevant.

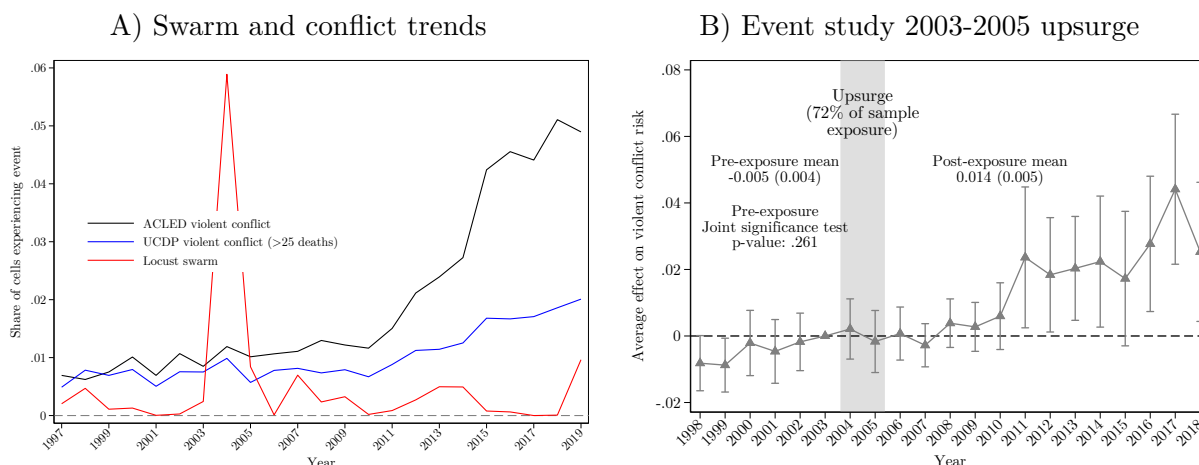
One possible explanation for delayed impacts is exposure to subsequent economic shocks. These could further lower the opportunity cost of fighting in swarm-exposed areas and increase violent conflict risk if the effect from the initial locust shock was not sufficiently severe. I test for this possibility by estimating heterogeneity in impacts of swarm exposure by whether a country is experiencing a famine and whether a cell is experiencing a severe drought in a given year (Table A4). Swarm exposure has a significantly larger impact on violent conflict in countries experiencing famine, but the average effect outside these contexts is similar to the overall average effect. On the other hand, there is no significant difference in the effect of swarm exposure by whether a cell experienced drought either in the current or previous year. I cannot test heterogeneity in impacts by subsequent exposure to locust swarms as control cells by definition have no exposure during the sample period, but I find no heterogeneity in impacts by exposure prior to 1990 which could have suggested either adaptation to or compounding effects of repeated swarm exposure.

These results indicate that the subsequent famine and drought shocks I examine do not explain the dynamic pattern of impacts of swarm exposure. Another possibility is that swarm exposure may create underlying conditions for violent conflict to occur locally after the onset of broader instability due to other factors. A central contribution of this paper is to consider the role of such conditional amplification of conflict in explaining the delayed effects of severe agricultural shocks.

5.1 Treatment timing and broader instability

An important observation is that the gap between swarm exposure and the largest impacts on violent conflict risk corresponds to the gap between the timing of the main swarm exposure event in the sample period—the 2003-2005 upsurge, accounting for 72% of sample exposure—and the years when the general risk of conflict increased across the sample countries due to the Arab Spring, the spread of Islamic terrorist groups, and multiple civil wars, as shown in Figure 4 Panel A. Panel B shows that exposure to this upsurge did not significantly increase the risk of violent conflict until 2011, the year of several uprisings related to the Arab Spring, but that effects remain large and statistically significant in subsequent years.²⁰ This does not mean that the average effect is mechanically driven by this upsurge. As discussed in subsection 4.3, average effects remain similar when cells first exposed in 2003–2005 are excluded and when country groups associated with major subsequent conflict increases are omitted. The dynamic pattern of exposure impacts excluding the 2003–2005 cohorts is noisier but also remains broadly similar.

Figure 4: Changes in conflict environment and impacts of exposure to 2003-2005 locust upsurge on violent conflict risk



Note: Panel A shows the share of sample cells experiencing any locust swarm or conflict event by year. Panel B shows results for an event study of exposure to the 2003-2005 desert locust upsurge, with 2003 as the reference period, with the same fixed effects and controls specified in Equation 1. The dependent variable is a dummy for any violent conflict event, and treatment is defined as any locust swarm observation from 2003-2005. The bars represent 95% confidence intervals using SEs clustered at the sub-national region level. Observations are grid cells approximately 28×28km by year.

The dynamic patterns in Figures 3 and 4 are consistent with locust shocks becoming conflict-relevant through interaction with the surrounding conflict environment rather than through a general immediate onset effect. Null short-term effects may reflect relatively peaceful conditions at the time of the main locust exposure events, limiting the feasibility of fighting and the potential net returns. The largest impacts of swarm exposure are realized in a period characterized by multiple popular uprisings, civil wars, and outbreaks of Islamic militancy. Significant increases 2-5 years after exposure in the main event study and when dropping cells first exposed in 2003-2005, compared

²⁰I find similar patterns when looking at effects of the 2003-2005 upsurge in individual countries with different events precipitating the spread of violent conflict (Figure B10).

to null effects over this period for the 2003-2005 upsurge, are consistent with the same conditional-amplification logic. Later swarm exposure events occur closer to or during periods when broader instability and mobilization environments made conflict more likely.

5.2 Conditional amplification of conflict incidence

To formally test whether past swarm exposure conditionally increases conflict incidence in situations of broader insecurity, I estimate heterogeneity in average long-term impacts of swarm exposure by various indicators of civil conflict or insecurity. The objective is to derive empirical implications that discipline interpretation of the timing and heterogeneity patterns above. These tests do not directly identify the actor-level pathways through which prior exposure becomes conflict-relevant, such as recruitment into armed groups, participation in local militias, or targeting by outside actors. They instead indicate whether the effects of prior exposure are concentrated in periods and places where broader insecurity makes these pathways more likely to operate.

At the cell-year level, I measure whether there are any concurrent violent conflict events in surrounding cells, either in the encompassing 1° cell, in the surrounding sub-national region, or in the rest of the country outside the surrounding sub-national region. These measures rely on the ACLED data and do not take into consideration any particular causes of the conflict.

At the country-year level, I identify a variety of situations which led to increases in civil conflict and insecurity in the following years during the sample period. For each of these, I define a country as being in an insecure situation in all years after the year in which the particular situation began. I first consider the Arab Spring specifically, which began in late 2010 or early 2011 in a number of sample countries. For example, civil conflict in Libya began to increase in 2011 with the start of the Arab Spring and then continued with the subsequent civil war. Egypt underwent a revolution in 2011 as part of the Arab Spring and experienced a coup d'état in 2013 and continued higher levels of insecurity. I then look at cases of revolutions or coups d'état and of civil wars and separatist movements. For example, the 2013 coup d'état in Egypt falls under the former category, and the civil war in Libya falls under the latter. Finally, I analyze the spread of violent attacks by (largely Islamist) terrorist organizations across the sample countries. For example, armed Islamist violence escalated in Burkina Faso starting in 2016, with many attacks by Al Qaeda and IS affiliates that established control over many rural parts of the country. Sudan and Somalia are categorized as subject to violence from Islamist terrorist organizations for the entirety of the sample period.

Table 3 shows that the long-term effect of swarm exposure is concentrated in settings already experiencing broader conflict. Past swarm exposure has no detectable effect on violent conflict in locations and periods with no violent conflict elsewhere around the cell, in the sub-national region, or elsewhere in the country, but exposed cells are highly likely to experience violent conflict when conflict is occurring in surrounding areas.

This pattern is more consistent with conditional amplification of conflict incidence than with a general onset effect of swarm exposure itself. One possibility is that swarm exposure could be causing conflict within the cell that spills over into the rest of the region or country, but this

Table 3: Heterogeneity in impacts of exposure to locust swarms on violent conflict risk by indicators of civil conflict

Outcome: Any violent conflict event	(1) Any violent conflict in surrounding 1 deg cell	(2) Any violent conflict in surrounding sub-region	(3) Any violent conflict in country outside sub-region	(4) Post-onset of Arab Spring in country	(5) Post-revolution or coup d'etat in country	(6) Post-civil war or separatist movement in country	(7) Post-onset of Islamic terror groups in country
Effect of exposure, No civil conflict	0.001 (0.002)	0.001 (0.001)	0.001 (0.001)	0.009*** (0.003)	0.008*** (0.002)	0.008*** (0.002)	0.010*** (0.003)
Effect of exposure, Any Any civil conflict	0.098*** (0.013)	0.038*** (0.008)	0.020*** (0.005)	0.036*** (0.011)	0.029*** (0.008)	0.040*** (0.011)	0.025*** (0.007)
Observations	301664	301664	301664	301664	301664	301664	301664
<i>p</i> -value, test of equality of coefficients	0.000	0.000	0.000	0.022	0.009	0.004	0.058
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: The table presents results from separate regressions where I estimate heterogeneous average long-term impacts of locust swarm exposure on the probability of any violent conflict event in a cell-year by indicators of civil conflict using the BJS estimator and built-in heterogeneity option in the Stata implementation of the estimator. The civil conflict indicators in columns 1-3 are cell-year level variables, defined by the presence of any violent conflict event in the ACLED database for the stated areas around but outside a particular cell. The indicators in columns 4-7 are variables defined at the country level based on the timing that Arab Spring uprisings, revolutions or coups d'etat, civil wars or separatist movements, or Islamic terror attacks began in the country that are coded as one for all years following the start of the conflict process. Countries not exposed to such conflicts are coded as 0 for all years. At the bottom of the table I include *p*-values for the test of equality of the coefficients representing effects by the presence of civil conflict. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

seems unlikely given limited evidence of conflict onset or spillovers from swarm exposure (Table B6, Figure 3). I also find similar heterogeneity when considering violent conflict in the country outside the cell's region, which is less likely to reflect spillovers from swarm exposure in the cell.

Turning to country-level indicators of exposure to civil conflict or insecurity, I find that average effects of swarm exposure are statistically and economically much larger in contexts experiencing a given type of civil conflict. The effect of swarm exposure outside of national civil conflict situations is about half the overall average impact (Table 1), and remains significant because countries not experiencing a given type of insecurity may be experiencing other types. Swarm exposure in countries affected by the Arab Spring protests and uprisings increases the likelihood of any violent conflict by 3.6pp in the periods after these social and political movements occurred. Exposure increases the likelihood of violent conflict by 2.9pp in countries having experienced a revolution or coup d'etat, by 4.0pp in countries having experienced a civil war or separatist movement, and 2.5pp in countries with active Islamic terror groups engaging in violent conflict.

The heterogeneity in locust swarm impacts by indicators of local civil conflict or insecurity could potentially be due to several factors. First, insecurity may further decrease local labor productivity, decreasing the opportunity cost of fighting. If effects of the initial swarm exposure are not sufficient to motivate affected individuals to choose to fight, an additional negative shock could provide the necessary push. While larger effects of swarm exposure in years when countries are experiencing a famine would also be consistent with the explanation, I do not find significant difference by whether a cell is experiencing a drought, when agricultural productivity may be significantly lower (Table A4). Second, areas exposed to locust swarms may be less able to defend themselves from attacks and therefore be targeted by armed groups when these become active.

The same mechanisms that would make exposed areas more vulnerable—persistent reductions in assets—would also make them less attractive targets, however, so it is unclear how the expected returns to predation in these areas would change.

The other possibility relates to the costs and returns of engaging in violent conflict, which may change when broader instability creates opportunities for mobilization or lowers the barriers to joining armed groups. Individuals in areas with lower opportunity costs of fighting following a severe prior agricultural shock may generally not find switching to fighting optimal as violent conflict is a costly and collective activity. Formation of armed groups and recruitment of fighters will be easier when social or political tensions, mobilization networks, or active conflict organizations reduce the social, emotional, and monetary costs of fighting. While swarm exposure may itself create lasting grievances,²¹ the local variation in exposure may prevent this from leading to general mobilization except when broader instability creates opportunities for conflict participation or additional potential returns to violence.

This heterogeneity in swarm impacts relates to other studies of heterogeneous effects of agricultural shocks on conflict risk. Buhaug et al. (2021) find that drought only causes the onset of civil conflict among ethnic groups experiencing recent political marginalization, an indicator of likely heightened grievances. Berman and Couttenier (2015) study the short-term effects of exogenous income shocks through international agricultural commodity prices on the risk of violent conflict in areas producing these commodities. They find that external income shocks act as threat multipliers affecting the “geography and intensity of ongoing conflicts” (p. 759), pointing to a likely opportunity cost mechanism. Similarly, Bazzi and Blattman (2014) find that commodity price shocks primarily affect conflict incidence and not conflict onset.

Taken together, the evidence in this section indicates that past swarm exposure becomes conflict-relevant primarily in periods of broader instability and therefore affects conflict incidence conditionally rather than independently causing conflict onset. I next present a conceptual framework of opportunity cost and conflict accounting for such conditional conflict amplification, test whether the estimated impacts of swarm exposure align with predictions from the framework, and discuss what remains unresolved.

6 Framework and evidence on mechanisms

The empirical results above establish three patterns that a framework for interpreting the dynamic effects of agricultural shocks on conflict must explain. First, locust swarm exposure does not lead to an average immediate increase in violent conflict risk. Second, effects emerge with delay. Third, the impacts of prior exposure are concentrated in periods and places where broader instability makes violent conflict more likely. These patterns suggest that a simple short-run opportunity cost mechanism is incomplete, but also that severe agricultural shocks may still become conflict-relevant

²¹For example, Narciso and Severgnini (2023) shows that individuals in families more affected by the Great Irish Famine in 1845-1850 were more likely to participate in the Irish Revolution in 1916-1921, and argues that impacts operate through persistent effects on grievances against British rule.

through longer-run and conditional channels.

Agricultural and weather shocks may affect conflict risk through many channels, including effects on production, physiology, psychology, infrastructure, institutions, and social relations (Burke et al., 2024; Dell et al., 2014; Hsiang and Burke, 2013; Mach et al., 2020; Mellon, 2022; Sarsons, 2015). For the purpose of interpreting the evidence, I focus on mechanisms operating through agricultural production and income while acknowledging that other channels may also matter.

The framework builds on models of agricultural shocks and intergroup conflict in which actors allocate labor between productive activities and fighting (Chassang and Padró i Miquel, 2009; Dal Bó and Dal Bó, 2011; McGuirk and Burke, 2020). These models emphasize how productivity shocks affect the opportunity cost of fighting, while also allowing shocks to affect the returns to fighting over output or production factors—the rapacity mechanism. I return to rapacity in interpreting the null short-term effects of swarm exposure, but otherwise focus on opportunity cost channels.

I present a simple occupational-choice framework that extends this logic in two ways. First, I add a dynamic component in which I allow a transitory shock to have persistent indirect effects through wealth or permanent income, lowering the future opportunity cost of fighting. Second, I make the broader instability or mobilization environment explicit: lower opportunity costs may increase conflict incidence primarily when conflict processes are already active. The objective is not to identify mechanisms definitively, but to derive empirical implications that discipline interpretation of the timing and heterogeneity patterns above. I then test whether swarm exposure affects economic outcomes consistent with a permanent income mechanism, before discussing what the evidence rules out, what it is most consistent with, and what remains unresolved.

6.1 Conceptual framework

In each period, individual i allocates one unit of labor to agricultural production, non-agricultural work, or violent conflict. Violent conflict is treated as a potential occupation with appropriable returns, while acknowledging that conflict may also have social, political, or psychological objectives. Returns to all activities depend on wealth W , where W broadly includes human, physical, and financial capital, and on individual and location characteristics X . I separately denote by M the broader instability or mobilization environment, such as the presence of organized armed groups, local insecurity, or proximate conflict processes, which affects the costs and returns to joining violent conflict. I treat M as given with respect to the individual’s occupation decision in period t ; the framework is not intended to explain the origins of this broader instability.

An adverse agricultural shock S reduces the returns to agricultural labor, with $\frac{\partial F^A}{\partial S} < 0$. This is the *short-run opportunity cost mechanism*: producers have less to lose by engaging in conflict. A shock may also reduce the output available to capture, partially offsetting this effect through the rapacity mechanism (Dube and Vargas, 2013). I return to this possibility below, but focus here on opportunity cost channels.

The key extension is to allow a transitory agricultural shock to have persistent indirect effects through wealth W . Costly coping responses after an income shock can reduce productive assets

and future productivity (de Janvry et al., 2006; Dercon, 2004; Dinkelman, 2017; Hallegatte et al., 2020; Hoddinott, 2006). I therefore allow current agricultural returns to depend on wealth affected by past shocks, with $W_{i,t} = W_{i,t}(\{S_{i,t-s}\}_{s=1}^{\tau})$ and $\frac{\partial F_t^A}{\partial S_{i,t-s}} < 0$.²² This captures the possibility that a transitory shock lowers future opportunity costs through a *permanent income mechanism*.²³

The potential net returns to fighting are $F^C(L_{i,t}^C, M_{i,t}, X_{i,t}, W_{i,t}, \{I_{j,t}, W_{j,t}, X_{j,t}, M_{j,t}\}_{j \in J})$, where J denotes potential targets. The broader instability environment $M_{i,t}$ can reduce the costs of fighting or increase expected returns by making organized armed groups, recruitment networks, or conflict opportunities more accessible. This is the core *conditional amplification* logic: reductions in the opportunity cost of fighting are more likely to increase conflict incidence when the broader instability environment lowers the threshold for participation. This follows the insight in Buhaug et al. (2021) that opportunity cost models can explain economic motives for conflict but not when these motives translate into action, while interpreting the relevant conditioning environment more broadly than grievances alone.

The individual's problem in period t can be written as:

$$\begin{aligned} \max_{L_{i,t}^A, L_{i,t}^N, L_{i,t}^C} \quad & I_{i,t} = F^A(L_{i,t}^A, S_{i,t}, W_{i,t}(\{S_{i,t-s}\}_{s=1}^{\tau}), X_{i,t}) + F^N(L_{i,t}^N, W_{i,t}(\{S_{i,t-s}\}_{s=1}^{\tau}), X_{i,t}) \\ & + F^C(L_{i,t}^C, M_{i,t}, X_{i,t}, W_{i,t}, \{I_{j,t}, W_{j,t}(\{S_{j,t-s}\}_{s=1}^{\tau}), X_{j,t}, M_{j,t}\}_{j \in J}) \\ \text{subject to} \quad & L_{i,t}^O \in \{0, 1\}, \quad F^O(0, \cdot) = 0, \quad \sum_O L_{i,t}^O = 1 \text{ for } O \in \{A, N, C\}. \end{aligned}$$

Individual i chooses violent conflict when the net returns from fighting exceed the highest returns from agricultural or non-agricultural work:

$$\begin{aligned} L_{i,t}^C = 1 \text{ iff } & F^C(1, M_{i,t}, X_{i,t}, W_{i,t}, \{I_{j,t}, W_{j,t}(\{S_{j,t-s}\}_{s=1}^{\tau}), X_{j,t}, M_{j,t}\}_{j \in J}) \\ & \geq \max \left\{ F^A(1, S_{i,t}, W_{i,t}(\{S_{i,t-s}\}_{s=1}^{\tau}), X_{i,t}), F^N(1, W_{i,t}(\{S_{i,t-s}\}_{s=1}^{\tau}), X_{i,t}) \right\}. \end{aligned}$$

In words, a negative agricultural shock can affect conflict risk immediately by lowering agricultural returns and opportunity costs of fighting, and later if coping responses reduce wealth and future productivity. But lower opportunity costs need not generate conflict in most periods. The instability environment $M_{i,t}$ changes the threshold at which lower returns to work translate into participation in violent conflict, allowing a past transitory shock to amplify conflict incidence when broader conflict processes are active.

6.2 Empirical implications and evidence

The framework yields four empirical implications for interpreting the results. First, a simple short-run opportunity cost mechanism predicts that a negative agricultural shock should increase conflict

²²I also allow $\frac{\partial F_t^N}{\partial S_{i,t-s}} < 0$ but expect this marginal effect to be smaller as the initial shock affects wealth through decreases in agricultural returns only.

²³The notation allows later shocks to compound the effects of initial exposure. In the empirical setting, repeated swarm exposure is rare, so I use first observed swarm exposure as the event-time anchor and interpret later outcomes as post-exposure dynamics following a severe transitory shock.

risk immediately after exposure by reducing the returns to agricultural production. This is the implication most directly emphasized in much of the literature on agricultural shocks and conflict. The results strongly reject this implication as a general pattern in this setting. As shown in [Figure 3](#), there is no average increase in violent conflict risk in the year of swarm exposure or the following year, when the direct negative shock to agricultural productivity should be strongest. This does not imply that opportunity costs are irrelevant, but it shows that a short-run opportunity cost mechanism alone cannot explain the dynamic effects of swarm exposure on violent conflict.

The model also allows for a countervailing rapacity channel. Lower agricultural output may reduce the returns to predatory attacks, potentially offsetting the effect of lower opportunity costs on conflict risk. If this explains the null immediate effect, the short-term response should be more attenuated for conflict over output than for conflict over factors of production, whose value is less directly affected by a transitory shock. Following McGuirk and Burke (2020), I proxy for output conflict using ACLED reports of violence against civilians, riots, and looting, and for factor conflict using UCDP violent conflict events. The comparison does not support rapacity as the explanation for the short-term null. Effects are close to zero in the year of exposure and the following year for both conflict types, and immediate point estimates are if anything slightly larger for output conflict than for factor conflict ([Figure A1](#)).²⁴ Decreased returns to predatory attacks therefore do not appear to explain the absence of immediate effects on violent conflict.

Second, a transitory agricultural shock may have long-term effects on conflict if it persistently reduces wealth, productivity, or outside options. In the framework above, this is formalized in the permanent income channel where $\frac{\partial F^A}{\partial S_{i,t-s}} < 0$. This implication is consistent with the significant long-term effects of swarm exposure in [Figure 3](#), but these effects alone do not identify the mechanism. The results show that effects can persist or emerge after the shock, but not why they do so.

Third, long-term opportunity cost effects should be most plausible where the original shock was agriculturally meaningful. This is supported by the evidence in [Table 2](#) and [Table 1](#): effects are concentrated in cells with agricultural land and among crop cells exposed during the main growing or harvest season. These results link the conflict effects to initial effects on agricultural production, making interpretations unrelated to agricultural destruction less plausible, but they do not show that persistent reductions in productivity or wealth following such destruction are the mechanism.

Fourth, lower opportunity costs may increase conflict incidence primarily when the broader instability or mobilization environment makes participation in conflict more feasible or attractive. The results in [section 5](#) provide strong evidence for this conditional pattern. Effects are largest when past swarm exposure overlaps with broader insecurity, including periods of civil conflict or other proximate conflict processes. This does not identify the origins of broader instability, show that the shock causes conflict onset, or imply that conditional effects are necessarily due to persistent decreases in opportunity costs. Instead, it shows that prior exposure becomes conflict-relevant

²⁴Later dynamic effects are generally positive and significant for both conflict types, though less precise for UCDP conflict. This is partly because UCDP conflict is less common ([Table A3](#) Panel A), but the average proportional effect is similar for both conflict types ([Table A3](#)).

primarily in environments where conflict processes are already active.²⁵ This implication also helps rationalize the rejection of the simple short-run opportunity cost prediction: a severe agricultural shock may lower opportunity costs, but the shock alone need not be sufficient to generate violent conflict.

Taken together, the evidence rules out a simple short-run opportunity cost mechanism as a complete explanation, provides little support for rapacity as the reason for the short-term null, and shows that the effects of past shocks are concentrated under broader instability. The remaining question is whether the long-term effects of swarm exposure are consistent with persistent economic disruption of the kind implied by the permanent income channel. I turn to this question by testing for effects of swarm exposure on measures of agricultural and economic activity.

6.3 Evidence related to persistent economic effects

Long-term increases in conflict risk could be explained by persistently reduced opportunity costs of fighting through the permanent income mechanism described above. The long-term conflict effects and agricultural specificity of the results are consistent with this interpretation, but they do not necessarily imply that swarm exposure persistently reduces productivity, wealth, or outside options. If this mechanism is important, we should expect at least some evidence of persistent effects on economic outcomes in exposed areas after the initial shock.

Severe immediate negative effects of locust swarms on agricultural output are well-documented (Green, 2022; Newsom et al., 2021; Showler, 2019; Symmons and Cressman, 2001; Thomson and Miers, 2002), and a growing body of household-level evidence finds persistent effects of swarm exposure on outcomes with longer-run economic implications. Most directly, Marending and Tripodi (2022) find that agricultural profits of exposed households in parts of Ethiopia are 20-48% lower two harvest seasons after swarm arrival. Other studies find lower educational attainment (De Vreyer et al., 2015; Yu et al., 2026), lower height-for-age (Conte et al., 2023; Gantois et al., 2024), and suggestive evidence of lower employment and salaries in early adulthood among children exposed while young (Yu et al., 2026).

Building on this household-level evidence, I test for average long-term effects of swarm exposure on measures of local economic activity at the grid-cell level in Table 4. I first consider agricultural productivity using annual maximum NDVI, remote-sensing estimates of major crop yields, and mean crop yield across DHS survey locations in each cell.

On average, locust swarm exposure has no significant long-term effect on maximum NDVI, remotely sensed crop yields, or DHS crop yields. The dynamic NDVI estimates are negative in the year of exposure and the following year, but similar in magnitude to differences before exposure and not persistent (Figure A2 Panel A). Estimated differences in remotely sensed crop yield are noisy and statistically insignificant before and after exposure (Figure A2 Panel B). These null average effects are consistent with locust swarms being temporary direct shocks to local agricultural

²⁵This paper does not claim to identify whether prior exposure matters through recruitment, local participation, targeting by armed groups, migration, grievances, trust, or other actor-level pathways.

Table 4: Average impacts of locust swarm exposure on indicators of economic activity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Max cell annual NDVI	Mean cell estimated main crop yield (kg/ha)	Cluster avg crop yield (kg/ha)	Cluster total crop prodn area (ha)	Cluster total crop prodn (metric tons)	Cluster tropical livestock units per km ²	Estimated net migration (per 1000 ppl)
Average effect of swarm exposure	-0.000 (0.002)	35.9 (37.7)	54.9 (78.7)	-116.7*** (44.1)	-316.7 (218.4)	-0.897 (2.304)	-4.691 (3.391)
Observations	258346	50144	3912	4075	4075	4075	249645
Outcome control mean	0.242	2694.7	3567.8	2308.6	12751.5	41.352	-0.494
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table presents results from separate regressions of swarm exposure on different outcomes, following Equation 1. Differences in sample sizes are due to differences in data availability across outcomes. Crop-specific yield estimates are only available in cells where that crop is produced. Data on DHS survey clusters are only available in cell-years with any completed DHS surveys within the cell. NDVI is calculated from MODIS satellite imagery (Didan, 2015) as the maximum of monthly average NDVI values in each cell for 2000-2018. Crop-specific yield data are from Cao et al. (2025) for 1997-2015, where the ‘main’ crop in cells with multiple crops is defined as the highest-yield crop in the cell. DHS cluster data are from the DHS ARcNA database for 1997-2018 (IFPRI 2020) and represent average values within DHS clusters at the time surveys were conducted. Annual net cell migration for 2000-2018 is from Niva et al. (2023). All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

production, rather than shocks that mechanically reduce productivity in later years. They also indicate that to the extent swarm exposure affects later labor productivity in affected areas, this is not reflected in measures of NDVI or crop yields at the grid-cell level.

While swarm exposure has no long-term effect on broad measures of agricultural productivity, it significantly decreases total crop area planted in DHS survey clusters. Cluster crop production area falls by 117 ha (5.0%) on average in the years following swarm exposure. While there is no significant effect on total cluster crop production quantity, the point estimate is large and negative. The decrease in crop production area could indicate a transition away from agriculture in exposed areas. I find no impact on livestock ownership, providing little direct support for persistent asset depletion in this sample.

Finally, I consider effects on migration, a common response to locust crop destruction (Ghorpade, 2024; The World Bank, 2020; Thomson and Miers, 2002). Using data from Niva et al. (2023), I find that swarm exposure does not significantly affect net annual migration in subsequent years. The point estimate indicates 4.7 more people per 1000 population migrating out of exposed areas per year on average, and increases in out-migration are significant in 3 post-treatment periods (Figure A2 Panel C). Increased out-migration could be consistent with lower local labor productivity or transitions away from agriculture, but the average effect is not precise enough to provide strong evidence for this channel.

One possible reason for the limited effects on most outcomes in Table 4 is that these intent-to-treat estimators capture only a diluted measure of treatment intensity. The median locust swarm covers around 50 km², or 6% of the area of a 0.25 degree cell. Many DHS clusters in exposed cells will likely not have experienced agricultural destruction, while average NDVI, remotely sensed crop yields, and net migration are measured over the entire cell. This grid-cell approach reduces measurement error in the conflict analysis but is less likely to capture economic impacts

concentrated in directly affected areas. More targeted analyses near swarm reports would be more likely to detect local economic effects, but would raise difficulties in defining exposed areas.

Taken together, a permanent income channel affecting the long-term opportunity cost of fighting remains plausible, especially in light of other evidence on persistent effects of locust exposure on economic outcomes, but the grid-cell evidence in this paper provides only limited and indirect support for this channel. The main mechanism may instead involve other persistent consequences of swarm exposure, or opportunity cost effects may matter only in combination with such channels. The next subsection discusses these possibilities.

6.4 Other potential long-term mechanisms

The evidence above is consistent with persistent economic disruption, but it does not uniquely identify a permanent income or opportunity cost mechanism. This matters because conditional amplification does not require that opportunity costs be the only or primary long-term channel through which swarm exposure affects conflict risk. The lack of immediate effects makes conflict hysteresis unlikely to be the primary driver of long-term impacts, despite the high autocorrelation of conflict over time. Prior exposure may instead, or additionally, affect future conflict incidence through other persistent changes in local economic, social, or political conditions.

Desert locust swarms are localized natural disasters with concentrated effects on agricultural production in only part of each 0.25° cell, potentially increasing local inequality or perceived relative deprivation (Bartusevičius, 2014; Østby, 2013). Households also typically view prevention, control, and relief as government responsibilities (Thomson and Miers, 2002), so limited support after exposure could increase grievances through perceived state failure or reduced trust in the state. These possibilities connect to models emphasizing grievances in civil conflict (Collier and Hoeffler, 2004) and to evidence that income shocks can interact with social or political tensions (Berman and Couttenier, 2015; Buhaug et al., 2021).

I do not directly test whether swarm exposure creates or deepens grievances, but this channel provides one way a transitory agricultural shock could have persistent effects on later conflict risk beyond the permanent income mechanism. More generally, such shocks may affect social attitudes, beliefs about future risk, or psychological responses to disaster exposure, all of which could influence the perceived costs or returns to participating in conflict. These alternative mechanisms are not mutually exclusive with the permanent income channel. A locust swarm could reduce wealth or productivity while also changing trust, social attitudes, perceived relative deprivation, or beliefs about future risks. The evidence in this paper does not separately identify these channels. Instead, the results show that severe agricultural shocks do not generate immediate conflict on average, but can have delayed effects that are concentrated when broader instability is present. Long-term economic disruption remains a plausible mechanism, but opportunity costs alone are unlikely to fully explain the dynamic and conditional pattern of effects.

7 Generalizability and implications for short-term designs

The framework in Section 6.1 is not specific to desert locust swarms: any severe transitory shock to agricultural production could have delayed effects on conflict risk if it persistently affects economic or social conditions and becomes conflict-relevant under broader instability. At the same time, locust swarms are a unique and catastrophic agricultural shock, raising the question of whether the dynamic and conditional patterns documented above generalize beyond this setting. I explore this question by comparing the impacts of locust swarm exposure to the impacts of drought exposure, a more commonly studied agricultural shock, using the same empirical framework. I then discuss the implications of the results for estimating impacts of transitory economic shocks that may nevertheless have persistent effects.

7.1 Comparing locust swarms and drought

Many studies find short-term impacts of drought exposure on conflict risk,²⁶ but I am not aware of any considering dynamic impacts over time. To measure drought exposure I follow several of these papers in using the Standardized Precipitation and Evapotranspiration Index (SPEI) which combines both precipitation and temperature-driven evapotranspiration pressure to capture drought conditions. The units of the SPEI are standard deviations from the historical average within a grid cell, where values between -1 and 1 are typically considered near normal conditions. I use monthly data from the SPEI Global Drought Monitor (Begueria et al., 2014) as compiled in the PRIO-GRID database. I define a cell as experiencing a drought shock in a particular year if there are at least 4 consecutive months where the SPEI is below -1.5, where this streak can include the last months of the previous year. This value is chosen to reduce the probability of multiple such drought exposures during the study period. A streak of at least 4 drought months is observed in 3.6% of all cell-years, compared to 7.7% for streaks of at least 3 months and 22.3% for streaks of at least 2 months.²⁷ As with locust swarm exposure I identify the first year during the sample period in which a cell experiences a drought and consider cells to be ‘affected’ in all subsequent years. Across all sample cells, 48.6% experience at least one drought from 1996-2014.

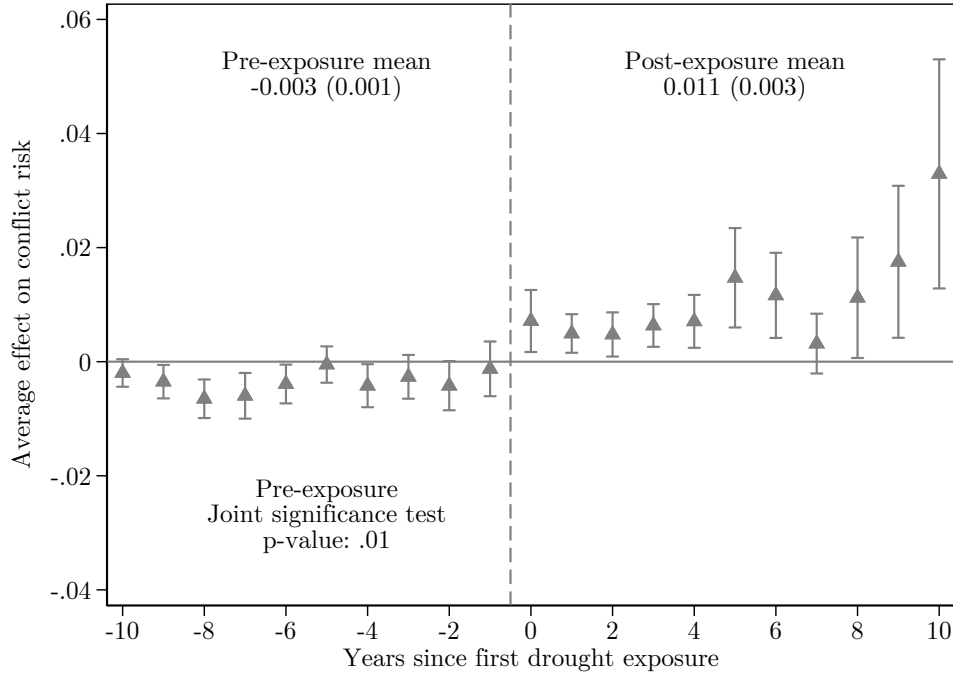
Figure 5 shows the results from an event study of drought exposure. Pre-exposure coefficients indicate a slightly lower but fairly stable baseline risk of violent conflict in areas ever exposed to drought compared to those not yet or never exposed. Treatment effect estimates are positive for all post-exposure periods and are statistically significant for 10 of 11 periods. Violent conflict risk increases by a statistically significant 0.7 percentage points in the year of drought exposure. This immediate effect contrasts with the null immediate response to desert locust swarms, and indicates that immediate conflict responses can arise for some agricultural shocks or contexts. The main

²⁶See e.g., Buhaug et al. (2021), Couttenier and Soubeyran (2014), Harari and La Ferrara (2018), Jia (2014), Maystadt and Ecker (2014), McGuirk and Nunn (2025), Von Uexkull (2014), and Von Uexkull et al. (2016)

²⁷Patterns of treatment effects are similar but standard errors are much larger if I use a threshold of 5 months but I find no significant effects with a threshold of 6 months (Table B8).

drought exposure event was in 2010,²⁸ around the time that insecurity and violent conflict in the study area began to increase (Figure 4 Panel A), which may help explain immediate impacts on violent conflict. A delay in the largest impacts of drought exposure on violent conflict nonetheless mirrors the pattern for locust swarm exposure; in the case of drought the largest effects occur 5 to 10 years after exposure.

Figure 5: Impacts of exposure to drought on violent conflict risk over time



The dependent variable is a dummy for any violent conflict event in a cell-year. The figure replicates Figure 3 but considering drought instead of locust swarm exposure. See the figure note for Figure 3 for more detail. Drought exposure is defined as ≥ 4 consecutive months in the year with $SPEI < -1.5$.

Table 5 compares average long-term impacts of swarm and drought exposure. Columns 1-3 reproduce results for impacts of swarm exposure from Table 1 and Table 3. Column 4 shows that on average, drought exposure increases the risk of any violent conflict in a given year by 0.8pp (73%).²⁹ Column 5 shows that impacts of drought on violent conflict are concentrated in cells with any crop land, with an even sharper difference than for effects of locust swarms. Column 6 tests for heterogeneity by activity of armed groups in the sub-region around exposed cells in a given year, an indicator of local insecurity. As with locust swarms, I find a precise null effect of drought exposure in cell-years with no violent conflict in surrounding cells in contrast to a large (1.8pp) increase in the probability of violent conflict in cell-years with such insecurity nearby. This heterogeneity relates to findings from Buhaug et al. (2021) and Michelini (2025) that the short-term impact of drought on conflict is driven by areas with marginalized ethnic groups (in Africa) and areas with

²⁸Nearly half (48.8%) of all exposure occurs in 2010 when around one-third of the study area was affected by drought, with no other year accounting for more than 8% of exposure.

²⁹This estimate is smaller than the average of the event study treatment period effects because treatment effects decrease more than 10 years after drought exposure, and are no longer statistically significant starting 13 years afterward (Figure B11).

higher predicted conflict risk outside of drought periods (globally), respectively.

Table 5: Average impacts of exposure to agricultural shocks on violent conflict risk

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
	Swarms	Swarms, Z = Any crop land	Swarms, Z = Any conflict in region outside cell	Drought	Drought, Z = Any crop land	Drought, Z = Any conflict in region outside cell
Average effect of shock exposure	0.018*** (0.004)			0.008*** (0.002)		
Effect of exposure, Z=0		0.007 (0.004)	0.001 (0.001)		-0.000 (0.001)	-0.000 (0.001)
Effect of exposure, Z=1		0.023*** (0.006)	0.038*** (0.008)		0.020*** (0.005)	0.018*** (0.003)
Observations	301664	301664	301664	452574	452559	452574
p-value, equality of effect by Z		.025	< .001		< .001	< .001
Outcome mean, no exposure	0.028	0.028	0.028	0.011	0.011	0.011
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: The table reproduces results for impacts of swarm exposure from Table 1 and Table 3, and then presents the same analyses considering impacts of drought rather than swarm exposure. See the table notes for Table 1 and Table 3 for more detail. Drought exposure is defined as ≥ 4 consecutive months in the year with SPEI < -1.5 . Observations are grid cells approximately 28 $\times$ 28km by year for 1997-2018 for swarms and 1997-2014 for drought. The sample for impacts of swarm exposure is restricted to cells within 100 km of a swarm observation.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The drought results suggest that the main insights of the paper are not unique to locust swarms. Another severe transitory agricultural shock also has conflict effects that vary over time and are larger in insecure environments. The heterogeneity by local insecurity in particular reinforces the idea that such shocks become conflict-relevant primarily through interaction with broader instability rather than by uniformly causing the onset of new conflict. The results also provide indirect reassurance that the large long-term impacts of swarm exposure on violent conflict are not driven only by selection in where locust swarms are reported. Drought exposure is measured using remotely sensed data and does not depend on local reports, while drought realizations over time are plausibly exogenous within cells because the drought index is based on deviations from cell-specific historical conditions. This is not a direct validation of the swarm design, and I interpret the qualitatively similar delayed and insecurity-contingent impacts of drought exposure as evidence on the generality of the dynamic and conditional patterns rather than as an independent validation of the locust identification strategy.

7.2 Implications for estimating short-term effects

These results have implications for empirical studies of economic shocks that may have persistent effects. This is particularly relevant for the economic shocks and conflict literature, where many studies of transitory shocks focus on immediate direct effects of the shocks and estimate only contemporaneous or short-term effects on conflict using distributed-lag two-way fixed effects models of the form:

$$Conflict_{ict} = \alpha + \beta_1 Shock_{ic,t} + \beta_2 Shock_{ic,t-1} + \delta X_{ict} + \gamma_{ct} + \mu_i + \epsilon_{ict} \quad (2)$$

This specification differs from the post-exposure specification in Equation 1, where the exposure

indicator tracks whether a cell has entered the period after first observed shock exposure and therefore allows conflict risk to change dynamically in all subsequent years. The distributed-lag specification instead assumes the outcome is unaffected in later years except as captured by the included lags (Burke et al., 2015). With cell fixed effects, short-term effects are estimated relative to other years in the same cell without a recorded shock, including later post-shock years. If shocks have persistent effects, this comparison can bias estimated short-term effects because later affected periods are treated as comparison periods.

Table 6 shows this pattern for estimated effects of locust swarms and drought on violent conflict. For locust swarms, a transitory-effects model estimates a highly significant 1.5pp *decrease* in violent conflict risk in the year of exposure.³⁰ The event study estimate for the same period is close to zero, while later effects are positive and the average long-term effect is 1.8pp. Including five lagged shock variables does not eliminate the bias: the year-zero estimate is 1.6pp lower than the event study estimate, and the estimates for the next five periods are consistently 1.4-2.0pp lower. For drought, the transitory-effects estimate for the year of exposure is a non-significant 0.4pp increase in conflict risk, compared to a statistically significant 0.7pp increase in the event study estimate. The differences are smaller than for locust swarms, consistent with the smaller average long-term effect of drought, but the transitory and five-year specifications still generally understate the dynamic effects.

The estimates in Table 6 show that imposing transitory effects can generate errors in both magnitude and sign of estimated short-term responses when shocks have persistent indirect effects on the outcome, affecting mechanism and policy interpretation. These results illustrate a special case of contamination in estimated effects of treatment leads and lags in settings with dynamic and heterogeneous treatment effects (Roth et al., 2023; Sun and Abraham, 2021). Simulations reported in Figure C3 and Table C1 provide intuition for this result: sign errors are most likely when true treatment effects in the shock period are small relative to effects in later periods, and adding lagged treatment terms attenuates bias under constant or decreasing long-term effects but worsens bias when treatment effects increase over time.

8 Conclusion

This paper shows that severe transitory shocks to agricultural production can have long-term dynamic effects on violent conflict risk, but not in the way implied by a simple short-run opportunity cost interpretation. Exposure to desert locust swarms significantly increases the long-term risk of violent conflict, yet there is no average increase in conflict in the year of exposure or the following year. Instead, effects emerge with delay and are concentrated when broader instability makes violence more likely, though this does not imply that swarm exposure causes the later conditions under

³⁰This estimated decrease is close to Tornngren Wartin (2018)’s estimate using similar data and a distributed-lag specification. While he interprets the result as suggesting endogenous under-reporting of locust swarm presence, the event study comparison suggests that the negative estimate can instead be explained by downward bias from ignoring long-term impacts of swarm exposure.

Table 6: Impacts of agricultural shocks on conflict risk, treating shock effects as temporary vs. persistent

	(1)	(2)	(3)	(4)	(5)	(6)
	Locust swarm			Drought		
	Transitory effects	5 year effects	Event study effects	Transitory effects	5 year effects	Event study effects
Any shock during year	-0.015*** (0.004)	-0.019*** (0.005)	-0.003 (0.003)	0.004 (0.003)	0.001 (0.002)	0.007** (0.003)
Any shock, 1 year lag		-0.012** (0.006)	0.005 (0.004)		-0.000 (0.002)	0.005*** (0.002)
Any shock, 2 year lag		-0.007 (0.006)	0.013*** (0.005)		-0.001 (0.002)	0.005** (0.002)
Any shock, 3 year lag		-0.008 (0.006)	0.012*** (0.004)		-0.002 (0.002)	0.006*** (0.002)
Any shock, 4 year lag		0.002 (0.007)	0.015*** (0.005)		-0.003 (0.003)	0.007*** (0.002)
Any shock, 5 year lag		-0.001 (0.007)	0.013** (0.005)		-0.004 (0.004)	0.015*** (0.004)
Average long-term effect			0.018*** (0.004)			0.008*** (0.002)
<i>p</i> -value, year 0 equality with event study	0.024	0.006		0.407	0.118	
<i>p</i> -value, year 5 equality with event study		0.090			0.002	
Observations	301664	233104	301664	454113	327746	452574
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Estimation	TWFE	TWFE	BJS	TWFE	TWFE	BJS

Note: The dependent variable is a dummy for any violent conflict event observed in a cell-year. All regressions estimate the effect of shock exposure on violent conflict controlling for total cell population and current year measures of total precipitation and maximum annual temperature along with cell and country-by-year fixed effects and distance to capital-by-year fixed effects. In columns 1 and 4 the shock is assumed to only have an effect in the year it is observed, and effects are estimated using TWFE. Columns 2 and 5 allow for persistent or delayed effects of the shock for up to 5 years after it is observed, and also include 5 lags of precipitation and temperature. Effects are estimated using TWFE. Columns 3 and 6 present a subset of the treatment effect estimates from the BJS event study estimator considering 10 pre-exposure and 10 post-exposure periods. These estimates are the same as those shown in Figure 3 and Figure 5. At the bottom of these columns I also present the estimates of average long-term effects of the shock from Table 5. At the bottom of columns 1, 2, 4, and 5 are *p*-values for tests of equality between coefficients under the transitory, 5 year, and event study models. Observations are grid cells approximately 28×28km by year for 1997-2018 for swarms and 1997-2014 for drought. The sample for impacts of swarm exposure is restricted to cells within 100 km of a swarm observation. SEs clustered at the sub-national region level are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

which effects on conflict risk are realized. The evidence therefore suggests that the relevant effect of severe agricultural shocks may be less about directly triggering conflict than about conditionally amplifying conflict risk once broader instability emerges or intensifies.

The empirical patterns discipline, but do not fully identify, the mechanisms. The results do not support the prediction that swarm exposure should on average immediately increase violent conflict risk by lowering the opportunity cost of fighting. I find little evidence that the null short-term effect is explained by offsetting changes in the returns to predatory conflict, and the absence of immediate effects also makes conflict hysteresis unlikely to explain the long-term increases. A permanent income channel, in which costly coping responses reduce later productivity and opportunity costs, remains plausible, especially given evidence from other studies showing long-term effects of swarm exposure on human capital and agricultural production. However, the evidence in this paper does not identify the actor-level pathway linking prior exposure to later conflict. Opportunity costs

may help explain why prior exposure to agricultural shocks matters, but they are unlikely to be sufficient on their own. Prior exposure becomes most conflict-relevant when the surrounding instability or mobilization environment makes violence more likely. Other persistent effects of swarm exposure—on inequality, trust, grievances, social attitudes, beliefs, migration, or targeting by armed actors—may also contribute to this delayed vulnerability.

These findings have broader implications for the climate-conflict and economic-shocks literatures. Similar patterns for drought exposure suggest that delayed and instability-contingent conflict responses are not merely an artifact of the locust setting, although the drought results are best interpreted as a complementary extension rather than an independent validation of the locust identification strategy. More generally, studies that define shocks as transitory and estimate only contemporaneous or short-term effects may miss important post-shock dynamics and bias estimated short-term effects when later post-shock years are treated as unaffected comparison periods. This can distort both mechanism interpretation and policy conclusions. Future work could test whether similar dynamic patterns appear for other climate, environmental, or economic shocks, and could help identify mechanisms by linking shock exposure to household or local measures of productivity, labor supply, wealth, inequality, trust, social attitudes, and psychological responses.

The results also suggest that policies mitigating the effects of agricultural shocks may reduce later conflict risk even when shocks do not immediately cause the onset of new conflict. Burke et al. (2024) find robust evidence across studies that higher living standards reduce the sensitivity of conflict risk to climate shocks. Consistent with this, existing evidence shows that social protection and insurance programs can reduce conflict risk directly or weaken the contemporaneous relationship between agricultural shocks and conflict (Croston et al., 2016; Fetzer, 2020; Sakketa et al., 2025). More generally, targeted protection and relief policies can protect human capital, accelerate local recovery, or mitigate the welfare effects of drought and disasters (e.g., de Janvry et al., 2006; Del Valle et al., 2020). An important next step is to test whether such policies also reduce the long-term conflict vulnerability documented in this paper, including by supporting recovery after the immediate production shock has passed. The conditional nature of the effects suggests that this support may be especially valuable when later political or security conditions deteriorate.

Finally, the results have implications for estimates of the economic and social costs of desert locust outbreaks and other agricultural disasters. Past research has debated whether costly desert locust monitoring and control operations generate net economic benefits, partly because outbreaks may have limited impacts on aggregate national measures of agricultural production (Joffe, 2001; Krall and Herok, 1997), while more recent work emphasizes large local damages, the value of proactive control, and the health benefits of preventing swarm exposure (Gantois et al., 2024; Showler, 2019; Zhang et al., 2019). The results here suggest that a broader accounting of the long-term economic and social impacts of agricultural destruction should include potential effects on violent conflict risk. More broadly, the possibility that severe agricultural shocks can affect conflict risk long after the direct shock has passed should be considered when weighing the costs and benefits of climate adaptation, disaster response, and agricultural resilience policies.

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A Additional figures and tables

Table A1: Summary statistics

Panel A: Yearly variables

	Mean	SD	Min	50 th	75 th	Max	N
Any violent conflict event - ACLED	0.02	0.14	0.0	0.0	0.0	1.0	557018
Any violent conflict event - UCDP	0.01	0.10	0.0	0.0	0.0	1.0	557018
Any swarms in cell	0.00	0.07	0.0	0.0	0.0	1.0	557018
Any swarms within 100km outside cell	0.05	0.21	0.0	0.0	0.0	1.0	557018
Any swarms within 100-250km of cell	0.11	0.32	0.0	0.0	0.0	1.0	557018
Population (10,000s)	1.63	8.92	0.0	0.1	0.9	749.8	557018
Total annual rainfall (100 mm)	2.40	3.79	0.0	0.8	2.8	43.4	557018
Max annual temperature (deg C)	37.55	5.11	11.5	38.2	41.3	49.1	557018

Panel B: Fixed variables

	Mean	SD	Min	50 th	75 th	Max	N
Any ACLED violent conflict event in cell from 1997-2018	0.13	0.33	0.0	0.0	0.0	1.0	25435
Any UCDP violent conflict event in cell from 1997-2018	0.07	0.26	0.0	0.0	0.0	1.0	25435
Any locust swarm in cell from 1985-2021	0.20	0.40	0.0	0.0	0.0	1.0	25435
Any locust swarm in cell from 1997-2018	0.09	0.29	0.0	0.0	0.0	1.0	25435
First exposed to locust swarm between 1997-2018	0.07	0.26	0.0	0.0	0.0	1.0	25435
First exposed to locust swarm in 2003-2005 upsurge	0.05	0.22	0.0	0.0	0.0	1.0	25435
Any locust swarm within 100 km from 1985-2021	0.78	0.41	0.0	1.0	1.0	1.0	25435
Any locust swarm within 100 km from 1997-2018	0.55	0.50	0.0	1.0	1.0	1.0	25435
Any cropland or pasture in cell	0.57	0.50	0.0	1.0	1.0	1.0	25435
Share of cell allocated to crops or pasture	0.23	0.32	0.0	0.0	0.4	1.0	25435
Any pasture in cell	0.56	0.50	0.0	1.0	1.0	1.0	25435
Share of cell allocated to pasture	0.18	0.27	0.0	0.0	0.3	1.0	25435
Any cropland in cell	0.31	0.46	0.0	0.0	1.0	1.0	25435
Share of cell allocated to crops	0.05	0.13	0.0	0.0	0.0	1.0	25435

Note: Observations are grid cells approximately 28×28km by year. The study period covers 1997-2018. Data on locust swarm observations is available from the FAO Locust Watch for 1985-2021. Data on conflict events are from the ACLED and UCDP databases. Data on population is from GPW (CIESIN, 2018). Data on precipitation and temperature are from WorldClim (Fick and Hijmans, 2017). Values for land cover are for the year 2000 from Ramankutty et al. (2010).

Table A2: Balance in cell characteristics by exposure to any locust swarm

	All cells		W/in 100km of any swarm		W/in 100km of any swarm	
	No weights		No weights		Inverse exposure propensity weights	
	Control	Treat	Control	Treat	Control	Treat
	Mean	Diff.	Mean	Diff.	Mean	Diff.
	(SD)	(SE)	(SD)	(SE)	(SD)	(SE)
Population in 2000 (10,000s)	1.22	1.32***	1.65	0.90***	1.67	0.43
	[6.99]	(0.37)	[8.67]	(0.33)	[8.71]	(0.50)
Mean of cell nightlights	0.05	0.01**	0.05	0.00	0.05	0.00
	[0.04]	(0.00)	[0.05]	(0.00)	[0.05]	(0.00)
Distance to national capital (km)	707.37	-182.42***	586.82	-61.87**	586.45	17.59
	[405.76]	(46.07)	[357.73]	(28.29)	[358.35]	(24.98)
Distance to nearest national boundary (km)	200.16	-40.97***	195.63	-36.45***	194.52	8.31
	[151.74]	(13.96)	[147.84]	(11.90)	[147.70]	(11.54)
Percent of cell covered by crop land in 2000	4.69	0.83	5.83	-0.31	5.89	0.63
	[13.12]	(0.78)	[14.48]	(0.69)	[14.55]	(0.80)
Percent of cell covered by pasture land in 2000	17.43	11.03***	21.59	6.87***	21.79	-0.98
	[26.69]	(2.40)	[28.05]	(2.16)	[28.12]	(1.59)
Percent of cell covered by barren area in 2009	68.97	-9.98***	61.16	-2.16	60.76	0.98
	[42.75]	(3.68)	[43.52]	(3.04)	[43.58]	(2.73)
Percent of cell covered by urban area in 2009	0.08	0.10	0.10	0.08	0.10	0.01
	[0.75]	(0.07)	[0.88]	(0.07)	[0.89]	(0.05)
Percent of cell covered by forest area in 2009	6.80	-1.09	7.12	-1.42	7.20	-0.07
	[17.00]	(1.03)	[16.15]	(0.91)	[16.22]	(0.92)
Percent of cell covered by water area in 2009	2.01	1.49**	2.92	0.58	2.94	0.20
	[10.70]	(0.61)	[12.75]	(0.61)	[12.81]	(0.53)
Mean annual max NDVI (1997-2018)	0.23	-0.01	0.24	-0.02	0.24	0.01
	[0.21]	(0.02)	[0.20]	(0.01)	[0.20]	(0.01)
Mean annual rainfall 1997-2018 (100 mm)	2.42	0.15	2.68	-0.11	2.71	0.11
	[3.80]	(0.25)	[3.76]	(0.17)	[3.77]	(0.18)
Mean annual max temperature 1997-2018 (deg C)	37.63	-1.61***	36.70	-0.69*	36.67	-0.21
	[5.11]	(0.45)	[5.05]	(0.37)	[5.07]	(0.38)
Mean of cell annual share of months with drought 1998-2014	0.09	-0.00	0.09	0.00	0.09	0.00
	[0.03]	(0.00)	[0.04]	(0.00)	[0.04]	(0.00)
Joint significance	$F = 5.00$ $p < 0.01$		$F = 2.38$ $p < 0.01$		$F = 0.64$ $p = 0.824$	

Note: The table shows results from separate bivariate regressions of baseline or mean cell characteristics on a dummy for being exposed to a locust swarm during the study period. The rows indicate which dependent variable is used. The first set of columns includes all cells while the second restricts the sample to cells within 100 km of any locust swarm observation from 1997-2021. The third set of columns includes all cells but weights observations by the inverse of the estimated propensity to have been exposed to a locust swarm during the study period. I include results of joint tests that there is no relationship between any of the characteristics and swarm exposure. Observations are grid cells approximately 28×28km by year. SEs clustered at the sub-national region level are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Average impacts of locust swarm exposure on different conflict types

	(1)	(2)	(3)
	Violent conflict (ACLED)	Output conflict (ACLED)	Factor conflict (UCDP)
Average effect of swarm exposure	0.018*** (0.004)	0.014*** (0.003)	0.006*** (0.002)
Observations	301664	301664	301664
Outcome mean post-2004, no exposure	0.028	0.019	0.012
Proportional effect of exposure	0.626	0.752	0.510
Country-Year FE	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes
Controls			

Note: Column 1 is the same as Table 1 column 1, and the other two columns replicate this for different conflict outcomes. The dependent variables are dummies for any conflict event being observed in a cell in a year. See the note for Table 1 for more detail on the estimation. Column 2 defines ‘output’ conflict as violence against civilians, riots, and looting from ACLED and column 3 defines ‘factor’ conflict as violent conflict events reported in the UCDP database (involving defined actors and at least 25 fatalities in a year), following McGuirk and Burke (2020).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

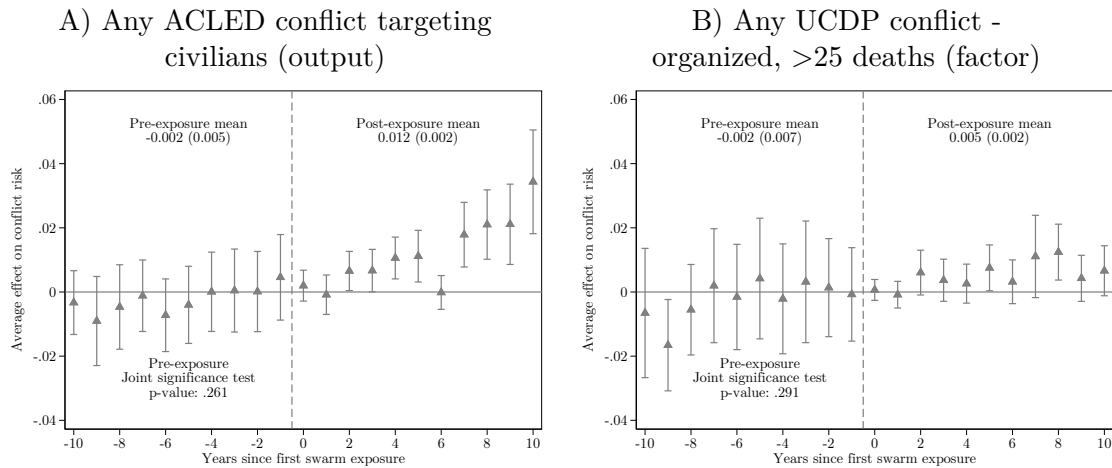
Table A4: Heterogeneity in impacts of exposure to locust swarms on violent conflict risk by exposure to other shocks

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)
	Any active famine in country	Any active drought in cell	Any drought prev. year in cell	Any swarm before 1990 in cell
Effect of exposure, No shock	0.016*** (0.004)	0.011*** (0.003)	0.018*** (0.004)	0.017*** (0.004)
Effect of exposure, Any shock	0.060*** (0.017)	0.015** (0.007)	0.023*** (0.008)	0.019*** (0.006)
Observations	301664	294353	295564	301664
<i>p</i> -value, test of equality of coefficients	0.010	0.511	0.527	0.626
Country-Year FE	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Note: The table presents results from separate regressions where I estimate heterogeneous average long-term impacts of locust swarm exposure on the probability of any violent conflict event in a cell-year by indicators of other shocks using the BJS estimator and built-in heterogeneity option in the Stata implementation of the estimator. The famine indicator in column 1 is defined at the country-year level by whether any famine was declared. The drought indicators in columns 2-3 are cell-year level variables, defined by whether there are at least 4 consecutive months where the SPEI is below -1.5. Swarm exposure prior to 1990 in column 4 considers exposure in the period from 1985-1989 where there were several major locust upsurges not considered in the definition of swarm exposure in the study sample period. At the bottom of the table I include *p*-values for the test of equality of the coefficients representing effects by the presence of civil conflict. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

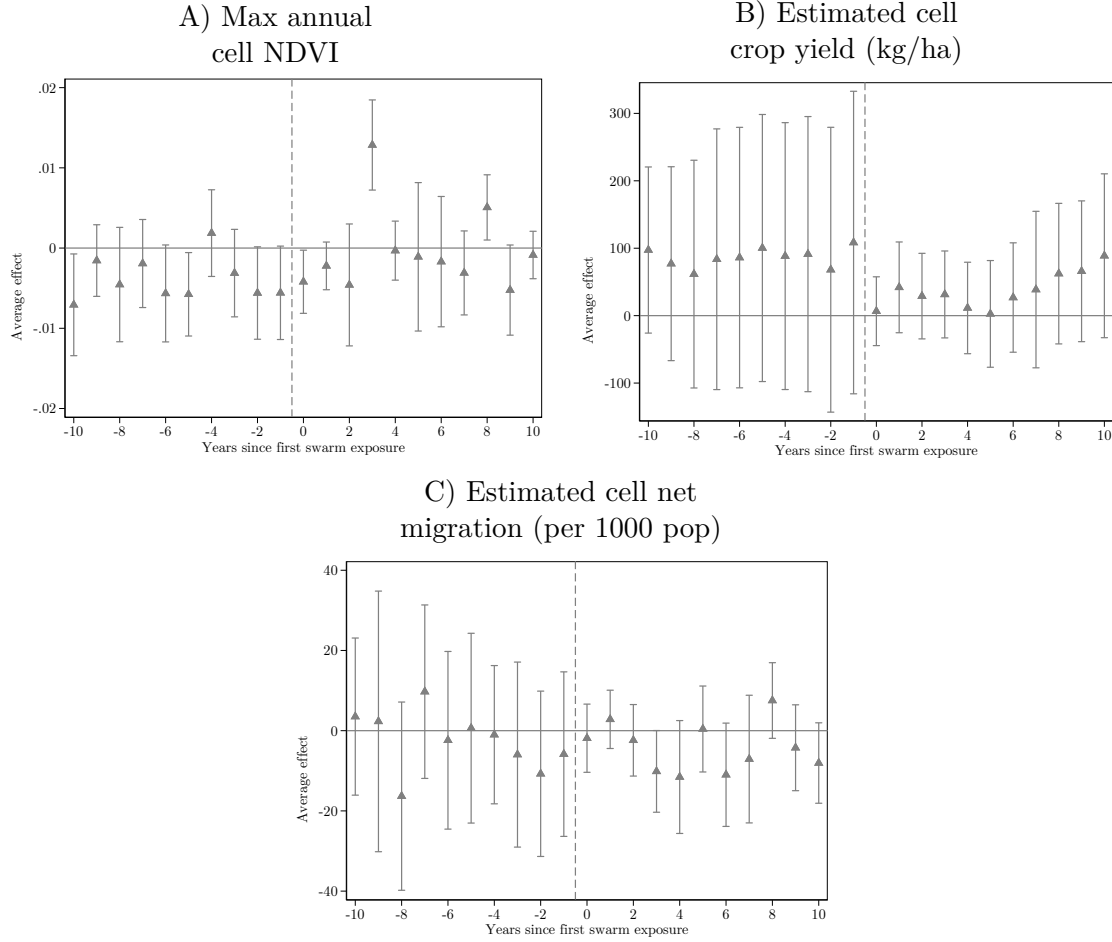
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A1: Impacts of swarm exposure on conflict risk over time, by conflict type



Note: The dependent variables are dummies for any conflict event being observed in a cell in a year, with the conflict type specified in the panel title. Each panel replicates Figure 3 for a different conflict outcome. See the figure note for Figure 3 for more detail. Exact coefficients are reported in Table B1. Average effects are reported in Table A3, and show that although the magnitude of effects on UCDP conflict are smaller, the proportional effects are similar as the mean incidence of UCDP conflict is much lower than for ACLED conflict.

Figure A2: Impacts of exposure to locust swarms on economic outcomes over time



Each panel replicates [Figure 3](#) but considering a different outcome as indicated in the panel title. See the figure note for [Figure 3](#) for more detail.

NDVI is calculated from MODIS satellite imagery from 2000-2018 (Didan, [2015](#)). I calculate average NDVI in each cell-month based on 16-day NDVI observations, and then take the maximum of these values in each cell-year. The event study is restricted to crop cells, where NDVI is a potential proxy for agricultural production. Crop-specific yield data at the cell level are estimated by Cao et al. ([2025](#)) for four main crops globally from 1997-2015. I consider the ‘main’ crop in cells with multiple crops as the highest-yield crop in the cell. Annual net cell migration for 2000-2018 is estimated by Niva et al. ([2023](#)).

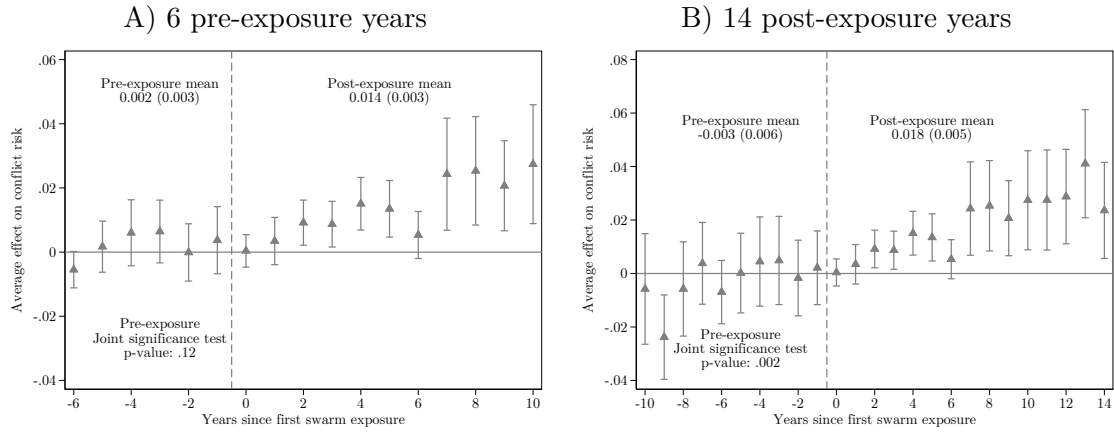
B Robustness

Table B1: Event study estimates of impact of swarm exposure on different conflict outcomes

	(1) Violent conflict (ACLED)	(2) Output conflict (ACLED)	(3) Factor conflict (UCDP)
pre10	-0.006 (0.011)	-0.003 (0.005)	-0.007 (0.010)
pre9	-0.024*** (0.008)	-0.009 (0.007)	-0.017** (0.007)
pre8	-0.006 (0.009)	-0.005 (0.007)	-0.006 (0.007)
pre7	0.004 (0.008)	-0.001 (0.006)	0.002 (0.009)
pre6	-0.007 (0.006)	-0.007 (0.006)	-0.002 (0.008)
pre5	0.000 (0.008)	-0.004 (0.006)	0.004 (0.010)
pre4	0.004 (0.009)	0.000 (0.006)	-0.002 (0.009)
pre3	0.005 (0.008)	0.000 (0.007)	0.003 (0.010)
pre2	-0.002 (0.007)	0.000 (0.006)	0.001 (0.008)
pre1	0.002 (0.007)	0.005 (0.007)	-0.001 (0.007)
tau0	0.000 (0.003)	0.002 (0.002)	0.001 (0.002)
tau1	0.003 (0.004)	-0.001 (0.003)	-0.001 (0.002)
tau2	0.009** (0.004)	0.007** (0.003)	0.006* (0.004)
tau3	0.009** (0.004)	0.007** (0.003)	0.004 (0.003)
tau4	0.015*** (0.004)	0.011*** (0.003)	0.003 (0.003)
tau5	0.013*** (0.004)	0.011*** (0.004)	0.008** (0.004)
tau6	0.005 (0.004)	-0.000 (0.003)	0.003 (0.003)
tau7	0.024*** (0.009)	0.018*** (0.005)	0.011* (0.007)
tau8	0.025*** (0.009)	0.021*** (0.006)	0.012*** (0.004)
tau9	0.021*** (0.007)	0.021*** (0.006)	0.004 (0.004)
tau10	0.027*** (0.009)	0.034*** (0.008)	0.007* (0.004)
Total annual precipitation (SDs)	0.004** (0.002)	0.002* (0.001)	-0.000 (0.001)
Max annual temperature (SDs)	0.012** (0.005)	0.005 (0.004)	0.005 (0.004)
Population (10,000s)	0.010*** (0.002)	0.009*** (0.002)	0.003*** (0.001)
Observations	295348	295348	295348

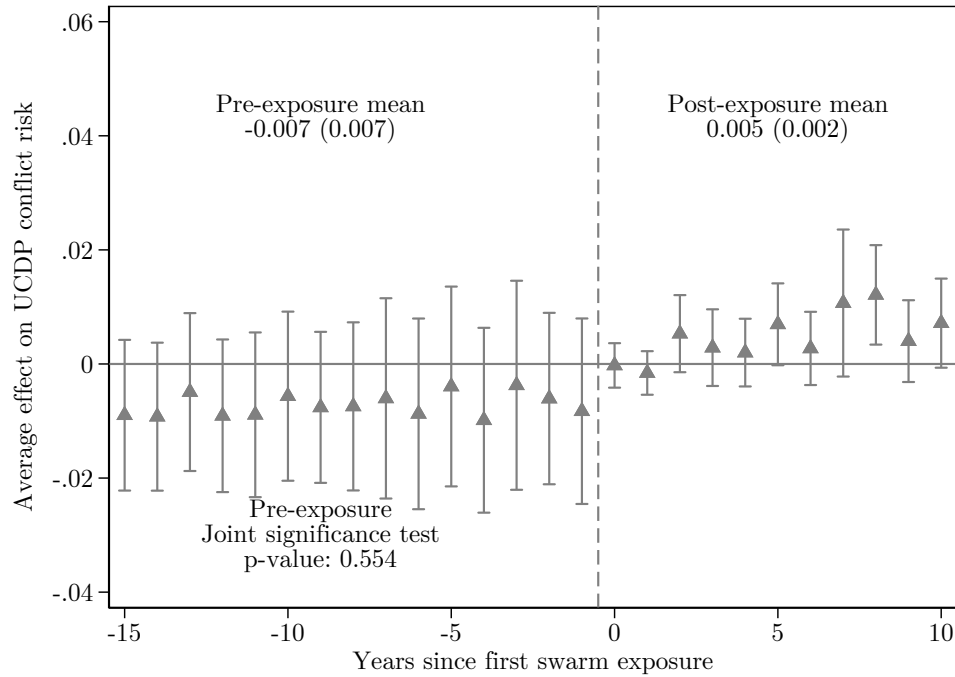
Note: The table shows the results of event study estimates of the impact of locust swarm exposure, with each column considering a different conflict outcome. Column 1 presents the results from [Figure 3](#), while columns 2 and 3 present the results from [Figure A1](#). Estimated impacts in each time period are weighted averages across effects for swarm exposure in particular years, calculated using the BJS estimator. Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital-by-year fixed effects. Observations are grid cells approximately 28×28km by year.

Figure B1: Impacts of exposure to locust swarms on violent conflict risk over different time horizons



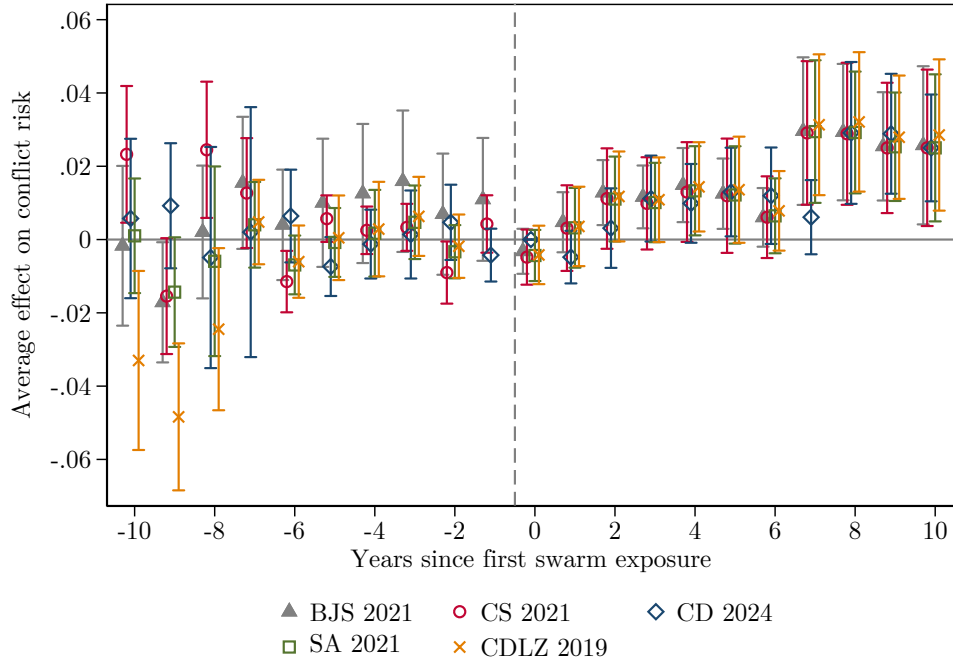
Each panel replicates [Figure 3](#) but changes the number of time periods included as indicated in the panel title. See the figure note for [Figure 3](#) for more detail.

Figure B2: Impacts of exposure to locust swarms on UCDP violent conflict risk with additional pre-exposure periods



This figure replicates [Figure 3](#) but considers UCDP violent conflict events instead of ACLED violent conflict events. It also changes the number of pre-exposure time periods, taking advantage of the long available time series of UCDP relative to ACLED data. [Figure A1](#) Panel B shows the results using the same number of pre-exposure time periods. UCDP events must have involved at least one organized actor and have resulted in at least 25 deaths in a calendar year. There are consequently fewer UCDP violent conflict events ([Figure 4](#) Panel A), so average effects of swarm exposure are smaller than for ACLED conflict but proportional effects are similar ([Table A3](#)).

Figure B3: Alternative locust swarm exposure event study estimators



The figure shows event study impacts of locust swarm exposure on the risk of any violent conflict estimated using different methods. ‘BJS 2021’ refers to the Borusyak et al. (2024) method first introduced in 2021, ‘CS 2021’ refers to Callaway and Sant’Anna (2021), ‘CD 2024’ refers to De Chaisemartin and d’Haultfoeuille (2024), ‘SA 2021’ refers to Sun and Abraham (2021), and ‘CDLZ 2019’ refers to Cengiz et al. (2019). Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. Observations are grid cells approximately 28×28km by year. All regression include cell and year fixed effects and no controls, in contrast to the main specification which uses BJS with cell and country-by-year fixed effects and controls, due to constraints in including these additional controls in the Stata packages implementing some of the alternative estimators. I test the sensitivity of the main BJS estimates to different fixed effects and controls in Figure B6. Table B2 shows the average effects by estimator.

Table B2: Average impacts of exposure to locust swarms on violent conflict risk by estimator

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
	BJS 2021	CS 2021	CD 2024	SA 2021	CDLZ 2019	TWFE
Average post-treatment effect	0.014*** (0.005)	0.014*** (0.005)	0.013*** (0.004)	0.015*** (0.004)	0.016*** (0.005)	0.017*** (0.006)
Observations	295348	301664	301664	301664	301664	301664

Note: The table shows the results from estimating average long-term impacts of swarm exposure on violent conflict using different estimators. ‘BJS 2021’ refers to the Borusyak et al. (2024) method first introduced in 2021, ‘CS 2021’ refers to Callaway and Sant’Anna (2021), ‘CD 2024’ refers to De Chaisemartin and d’Haultfoeuille (2024), ‘SA 2021’ refers to Sun and Abraham (2021), ‘CDLZ 2019’ refers to Cengiz et al. (2019), and ‘TWFE’ refers to the two-way fixed effects specification shown in Equation 1. As in Figure B3, all regression include cell and year fixed effects and no controls, in contrast to the main specification which uses BJS with cell and country-by-year fixed effects and controls, due to constraints in including these additional controls in the Stata packages implementing some of the alternative estimators. The results in column 1 using BJS and column 6 using TWFE therefore differ from those in column 1 of Table 1 and Table B3 which follow the main specification. The purpose of the table is to test sensitivity of the estimates to the choice of estimator. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

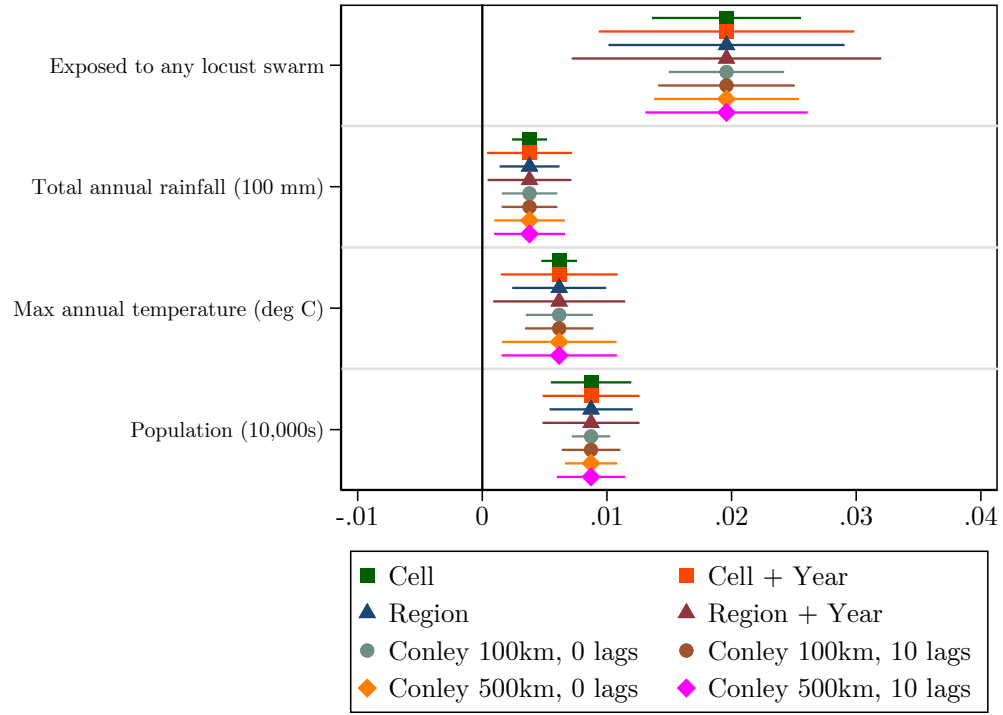
Table B3: TWFE average impacts of exposure to locust swarms on violent conflict risk by land cover

Outcome: Any violent conflict event	(1)	(2)	(3)
	All land	Land = Any crop or pasture land	Land = Any crop land
Exposed to any locust swarm	0.020*** (0.005)	0.005 (0.006)	0.006 (0.006)
Total annual precipitation (SDs)	0.004** (0.002)	0.002* (0.001)	0.002* (0.001)
Max annual temperature (SDs)	0.013** (0.005)	0.012** (0.006)	0.013** (0.006)
Population (10,000s)	0.009*** (0.002)	0.026** (0.011)	0.007 (0.006)
Exposed to any locust swarm × Land=1		0.017** (0.008)	0.021** (0.009)
Total annual precipitation (SDs) × Land=1		0.002 (0.002)	0.003* (0.002)
Max annual temperature (SDs) × Land=1		0.001 (0.004)	0.000 (0.005)
Population (10,000s) × Land=1		-0.018 (0.011)	0.002 (0.006)
Observations	304612	304612	304612
Outcome mean post-2004, no exposure	0.028	0.028	0.028
Country-Year FE	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Note: The table replicates the results in Table 1 but using a two-way fixed effects (TWFE) estimator. Column 1 includes no land cover interactions and the other two interact all right-hand side variables (except cell fixed effects) with cell land cover dummies. The ‘Land=1’ rows show the coefficients for the interaction of right-hand side variables with cell land cover dummies indicated in the column heading. The outcome mean for control cells is shown for post-2004 for comparison with exposure impacts in the period after the majority of swarm exposure occurred. All regressions include country-by-year, cell, and distance to capital by year fixed effects in addition to the controls shown. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure B4: Estimated coefficients from Equation 1 with different SEs



Note: The outcome variable is a dummy for any violent conflict observed. The figure shows 95% confidence intervals for estimates from Table B3 column (1) applying different clustering for the SEs. Regions are clusters of sub-national administrative units constructed so that each one includes at least 32 grid cells. Observations are grid cells approximately 28×28km by year. Regressions also include country-by-year and cell FE. Results for country-level clustering and wild bootstrap p-values are shown in Table B4.

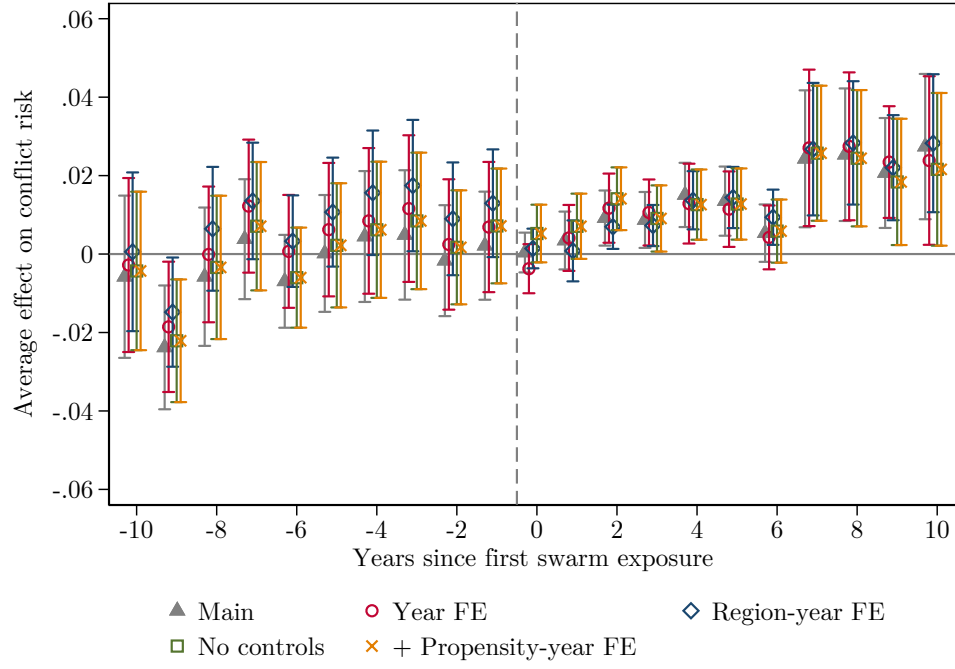
Table B4: TWFE average impacts by level of SE clustering

	(1) Region cluster	(2) Country cluster
Exposed to any locust swarm	0.020*** (0.005)	0.020*** (0.006)
Total annual precipitation (SDs)	0.004** (0.002)	0.004 (0.003)
Max annual temperature (SDs)	0.013** (0.005)	0.013 (0.010)
Population (10,000s)	0.009*** (0.002)	0.009*** (0.001)
Observations	304612	304612
Conventional p-value (treated)	0.000	0.004
Wild bootstrap p-value (treated)	.	0.000

Note: Column 1 replicates the results in [Table B3](#) column 1, while column 2 clusters standard errors at the country level instead of the sub-national region level. Because the sample includes only 23 countries, I also include the p-value from 10,000 wild-cluster bootstrap estimates.

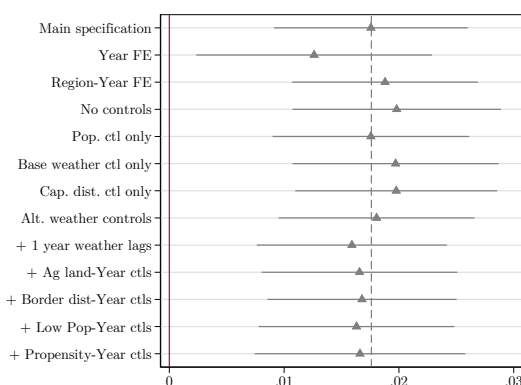
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure B5: Sensitivity of locust swarm exposure event study results to different specifications



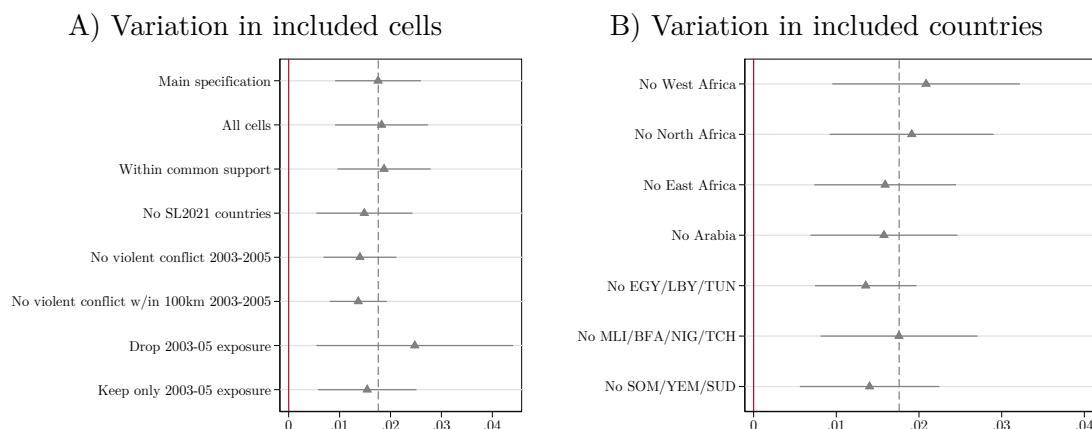
Note: Each event study replicates Figure 3 but changes some aspect of the specification as indicated in the legend. Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. Observations are grid cells approximately 28×28km by year. All regressions estimate dynamic effects of swarm exposure on violent conflict risk using the BJS estimator, and include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects unless otherwise stated. The year and region-year FE specifications replace country-by-year FE with FE at these levels, where region indicates the same sub-national regions used in clustering the SEs. The propensity-year FE specification adds estimated swarm exposure propensity by year fixed effects; the estimation of these propensities is explained in the Empirical approach section.

Figure B6: Sensitivity of average impacts of locust swarm exposure on violent conflict risk to alternative specifications



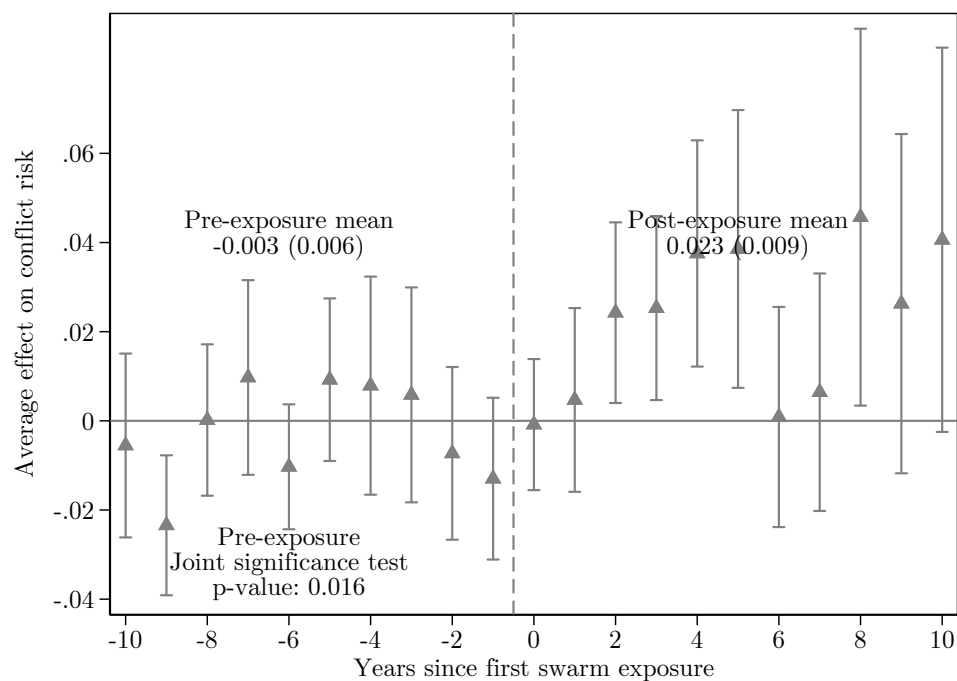
Note: Each estimate and 95% confidence interval is from a separate regression replicating Table 1 column 1 with a given modification. The dashed gray line indicates the main estimate from Table 1 column 1. The main specification includes controls for precipitation, temperature, and population at the cell-year level and cell, country-by-year, and distance to capital-by-year FE. Each coefficient is associated with a specific change in this specification. Region-year FE replace country by year FE with subnational region by year FE. ‘Alt.’ weather controls replace the rainfall and temperature measures from WorldClim with measures from CHIRPS and ERA5, respectively. The bottom five specifications add additional controls: 1 year lags of precipitation and temperature, any agricultural land-by-year FE, below median distance to border-by-year FE, below median baseline population-by-year FE, and estimated swarm exposure propensity-by-year FE.

Figure B7: Sensitivity of average impacts of locust swarm exposure on violent conflict risk to alternative samples



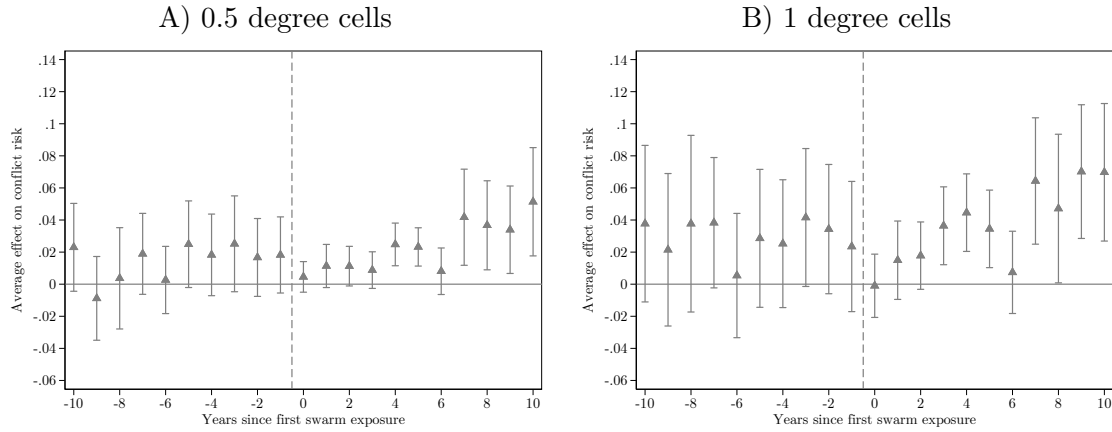
Note: Each estimate and 95% confidence interval is from a separate regression replicating Table 1 column 1 with a given modification. The dashed gray line indicates the main estimate from Table 1 column 1. The main specification excludes cells more than 100 km from any swarm report and outside the range of common support of estimated swarm exposure probability. Panel A shows results varying the set of included cells. I first vary whether cells more than 100 km from any swarm report and outside the range of common support of estimated swarm exposure probability are excluded. I then maintain the main sample restrictions but add additional restrictions. “No SL2021 countries” drops cells in countries where Showler and Lecoq (2021) report insecurity prevented some locust control operations during the sample period. “Drop 2003–05 exposure” excludes treated cells first exposed to locust swarms in 2003–2005. “Keep only 2003–05 exposure” retains treated cells first exposed in 2003–2005 but drops treated cells first exposed in other years. Panel B shows results from dropping countries in particular geographic regions or that experienced particular forms of violent conflict over the study period, to test whether results may be driven by an association of swarm exposure and unrelated later conflict in certain countries.

Figure B8: Impacts of locust swarm exposure over time, dropping cells first exposed in 2003-2005



Note: The figure replicates [Figure 3](#) but dropping all treated cells that were first exposed during the 2003-2005 locust upsurge. Never-treated cells are retained as comparison cells, and treated cells first exposed in other years remain in the analysis.

Figure B9: Dynamic impacts of swarm exposure on violent conflict risk at different scales



Each panel replicates [Figure 3](#) estimating dynamic impacts of locust swarm exposure on the risk of any violent conflict event using the BJS method at different spatial scales. The main analysis uses 0.25° cells. When collapsing to larger cells I take the maximum of the swarm exposure and violent conflict variables and the mean of control variables across smaller cells within the aggregate cell. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. SEs are clustered at the sub-national region level, which is assigned based on the region of the majority of 0.25° component cells. [Table B5](#) shows average effects.

Table B5: Average impacts of swarm exposure on violent conflict risk at different scales

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
Average effect of swarm exposure	0.018*** (0.004)	0.025*** (0.007)	0.028*** (0.011)	0.131*** (0.032)	0.118*** (0.033)	0.096*** (0.037)
Observations	301664	93940	30763	301664	93940	30763
Outcome mean post-2004, no exposure	0.028	0.067	0.128	0.028	0.067	0.128
Proportional effect of exposure	0.623	0.368	0.216			
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cell size (degrees)	0.25	0.5	1	0.25	0.5	1
Violent conflict outcome	Any	Any	Any	SDs	SDs	SDs

Note: The table replicates the results in [Table 1](#) column 1 at different spatial scales, as presented in the note to [Figure B9](#). Columns 1-3 show absolute effects on a dummy for any violent conflict while columns 4-6 show relative effects on the standard deviation in the probability of any violent conflict. The latter approach maintains comparability in effect sizes as the baseline risk of any conflict in a cell-year increase with cell size.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B6: Average impacts of direct and spillover exposure to locust swarms on violent conflict risk

Outcome: Any violent conflict event	(1) All swarms	(2) 2003-2005 upsurge swarms
Exposed to swarm	0.020*** (0.005)	
Exposed to swarm w/in 100km outside cell	0.001 (0.003)	
Exposed to swarm		0.018*** (0.005)
Exposed to swarm w/in 100km outside cell		0.011* (0.006)
Observations	301664	301664
Country-Year FE	Yes	Yes
Cell FE	Yes	Yes
Controls	Yes	Yes

Note: The table presents results from estimating Equation 1 but including a spillover swarm exposure variable based on the first year a swarm is outside the cell but within 100 km of a cell centroid. As with within-cell swarm exposure, spillover swarm exposure is coded as one in all years after the first nearby exposure in the study period. Column 1 considers all swarm exposure events while column 2 focuses on the 2003-2005 upsurge. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. SEs are clustered at the sub-national region level. Observations are grid cells approximately 28×28km by year.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

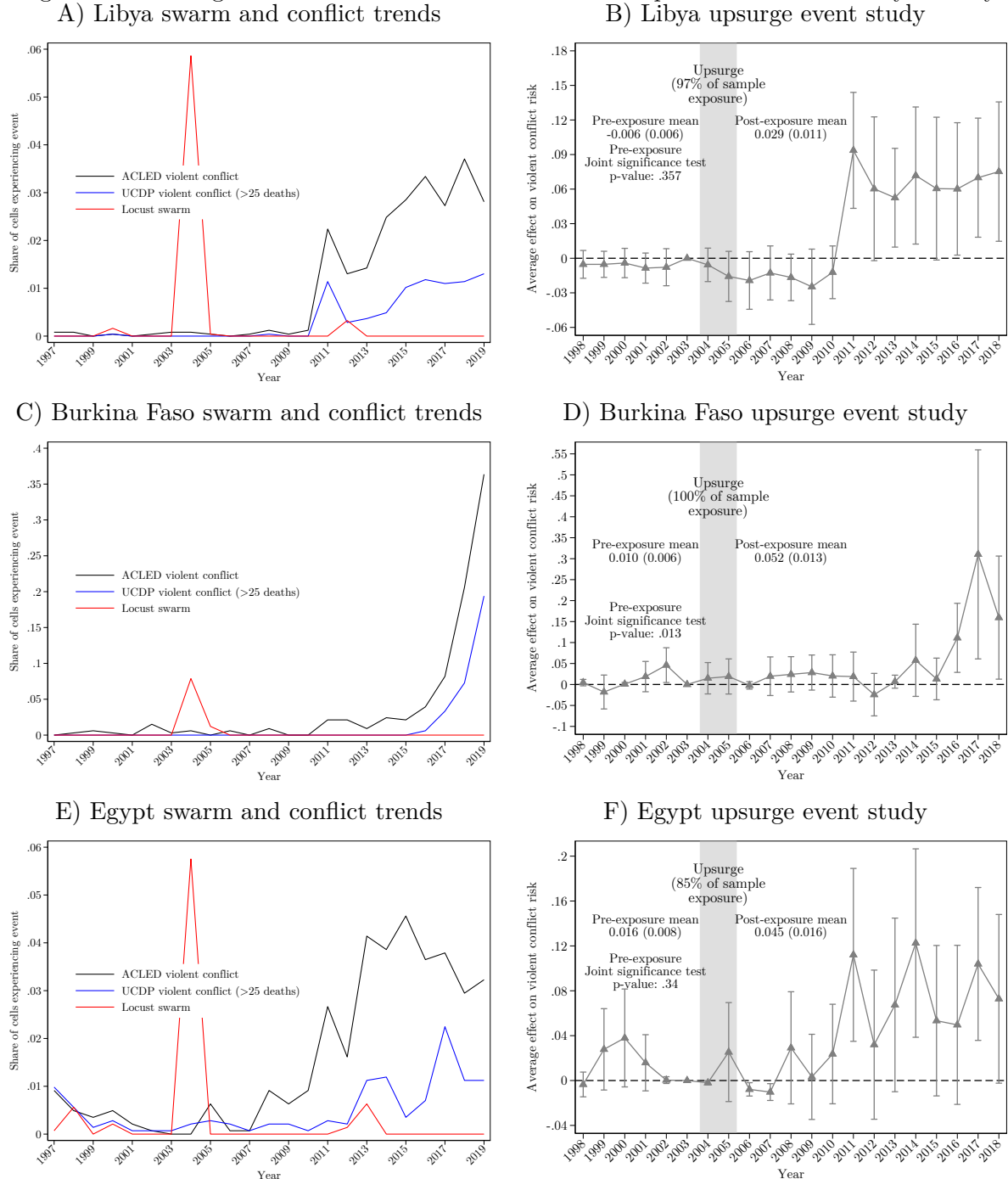
Table B7: TWFE average impacts by time since any and most recent exposure event

Window	Any exposure in window	Most recent exposure window
0–2 years	0.0045 (0.0035)	0.0064* (0.0034)
3–6 years	0.0129*** (0.0046)	0.0155*** (0.0045)
7–10 years	0.0281*** (0.0083)	0.0309*** (0.0084)
11+ years	0.0332*** (0.0088)	0.0340*** (0.0092)
Total exposed cells	1833	1833
Exposed cells with >1 exposure year	172	172

Note: These analyses use the main TWFE specification from [Table B3](#) column 1 but replace the dummy for being in a period after first swarm exposure with indicators for time since swarm exposure. In column 1, treatment is defined based on all swarm exposure events, so a cell-year can contribute to more than one exposure window. In column 2, treatment is defined based on the most recent swarm exposure event (as opposed to the first during the sample period). At the bottom, I show that only 172 of 1833 exposed cells are exposed to a swarm in more than one sample year, implying that the different definitions (first swarm exposure, any exposure, most recent exposure) will be identical for most exposed cells.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure B10: Changes in conflict environment and locust exposure event studies by country



Note: The figure replicates [Figure 4](#) for Libya, Burkina Faso, and Egypt alone, which were selected because of the strong concentration of locust exposure during the 2003-2005 upsurge. Conflict in Libya began to increase in 2011 with the start of the Arab Spring and the subsequent civil war. Armed Islamist violence escalated in Burkina Faso starting in 2016, with many attacks by Al Qaeda and IS affiliates that established control over many rural parts of the country. Egypt underwent a revolution in 2011 as part of the Arab Spring and experienced a coup d'état in 2013 and continued higher levels of insecurity.

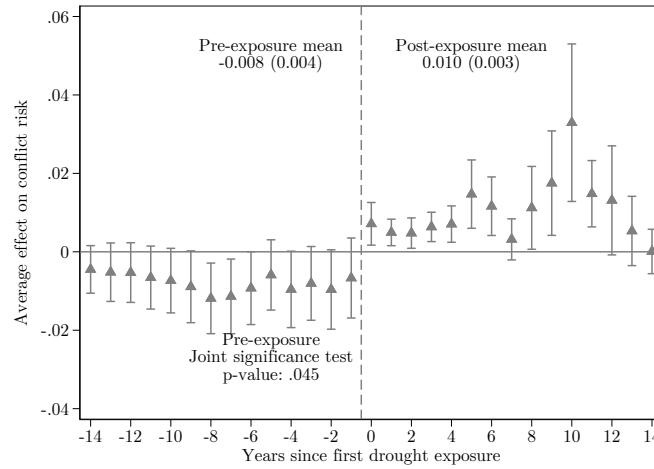
Table B8: Robustness of drought estimates to changing number of consecutive drought months

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
	5 month drought	5 month, Z = Any crop or pasture land	5 month, Z = Any conflict in country outside sub-region	6 month drought	6 month, Z = Any crop or pasture land	6 month, Z = Any conflict in country outside sub-region
Average effect of shock exposure	0.004** (0.002)			0.001 (0.002)		
Effect of exposure, Z=0		-0.002 (0.002)	0.000 (0.001)		-0.003* (0.002)	0.001 (0.001)
Effect of exposure, Z=1		0.008*** (0.003)	0.005** (0.003)		0.004 (0.003)	0.001 (0.002)
Observations	454986	454971	454986	455310	455300	455310
Outcome mean, no exposure	0.012	0.012	0.012	0.012	0.012	0.012
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: The table reproduces columns 4-6 of [Table 5](#) changing the definition of what constitutes a drought. The main definition uses at least 4 consecutive months of drought (observed in 3.6% of cell-years), and the alternative definitions presented here use thresholds of 5 months (1.8% of cell-years) and 6 months (1.1%).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

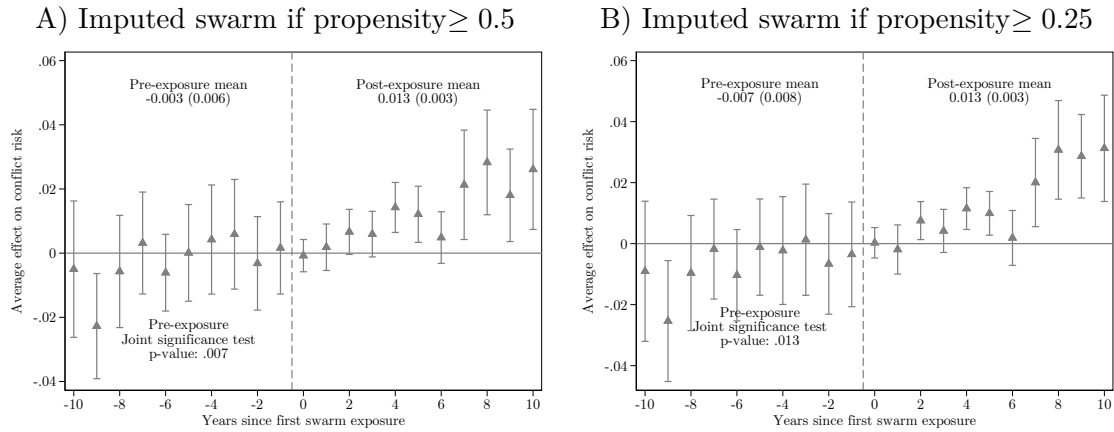
Figure B11: Event study impacts of drought exposure with 14 years of pre- and post-exposure estimates



This figure replicates [Figure 5](#) but changes the number of time periods included as indicated in the panel title.

C Simulations

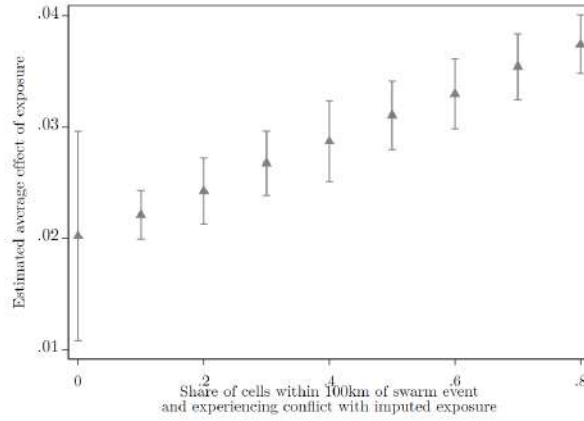
Figure C1: Impacts of exposure to locust swarms on violent conflict risk over time, imputing swarms in high-propensity areas



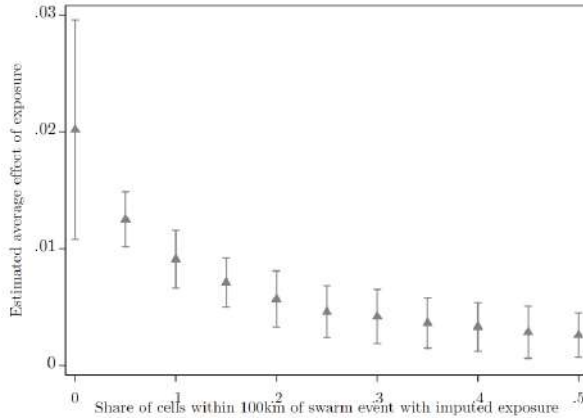
Each panel replicates [Figure 3](#) but changes the imputation of potential ‘missing’ locust swarms. In each case, I assign a cell as experiencing a locust swarm if its estimated propensity of exposure (over the whole study period) is above a certain level in any year where a locust swarm was reported within 100 km of the cell. I then define exposure based on the first year such a swarm is imputed, and estimate the event study as previously. See the figure note for [Figure 3](#) for more detail on the estimation.

Figure C2: Simulations of missing swarm observations

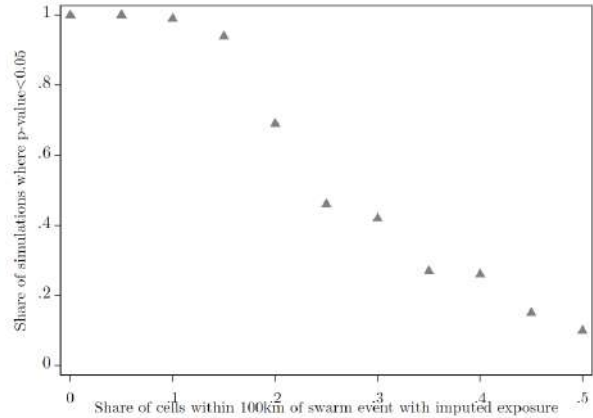
A) Betas, simulated swarms
in areas experiencing conflict
near swarm report



B) Betas, simulated swarms
in any area near swarm report



C) Rejection of H_0 , simulated swarms
in any area near swarm report



Note: The figures show the results from estimating Equation 1 in simulations imputing the presence of increasing shares of unreported locust swarms in cell-years with a locust swarm reported within 100 km. For each share, I run 100 simulations randomizing which cell-years are imputed with an unreported swarm, recalculating the swarm exposure treatment variable, and estimating the average long-term impact of swarm exposure on violent conflict risk. In Panel A, I only impute swarms in cell-years both experiencing violent conflict and within 100 km of a reported locust swarm, to simulate effects of missing swarm reports in insecure areas. In Panels B and C, I impute swarms across all cell-years within 100 km of a reported locust swarm. Panels A and B report the average estimated effect across all simulations by share of imputed swarms, along with a 95% confidence interval for these estimates. Panel C reports the share of simulations where the p -value for the coefficient on swarm exposure is less than 0.05, by share of imputed swarms.

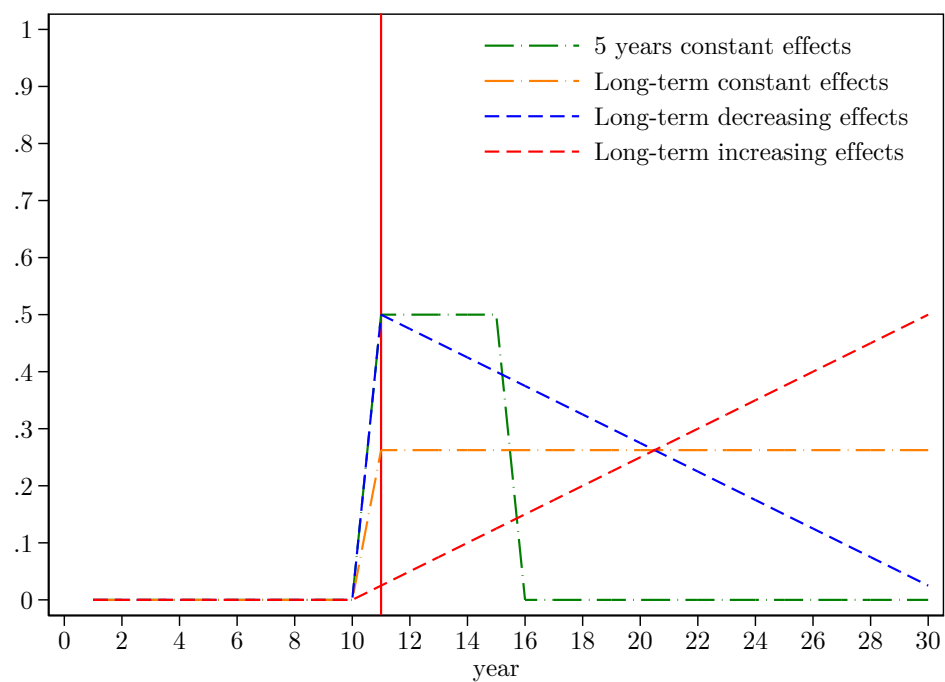
Table C1: Simulation: bias in estimates of short-term effects of shocks under different dynamic treatment effects

	(1) Immediate effect only	(2) Constant 5 year effects	(3) Constant long-term effects	(4) Decreasing long-term effects	(5) Increasing long-term effects
<i>A. True effects:</i>					
Treatment during current year	0.500*** (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.500*** (0.000)	0.025*** (0.000)
Treatment 1 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.475*** (0.000)	0.050*** (0.000)
Treatment 2 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.450*** (0.000)	0.075*** (0.000)
Treatment 3 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.425*** (0.000)	0.100*** (0.000)
Treatment 4 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.400*** (0.000)	0.125*** (0.000)
<i>B. Estimated transitory effects:</i>					
Treatment during current year	0.500*** (0.000)	0.431*** (0.000)	0.091*** (0.000)	0.336*** (0.000)	-0.155*** (0.000)
Difference from actual treatment effect	0.000	0.069	0.171	0.164	0.180
<i>C. Transitory effects w/ 1st differences:</i>					
Treatment during current year	0.500*** (0.000)	0.250*** (0.000)	0.131*** (0.000)	0.263*** (0.000)	0 (.)
Difference from actual treatment effect	0.000	0.250	0.131	0.237	0.025
<i>D. Estimated lagged effects:</i>					
Treatment during current year	0.500*** (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.380*** (0.000)	-0.170*** (0.000)
Treatment 1 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.355*** (0.000)	-0.145*** (0.000)
Treatment 2 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.330*** (0.000)	-0.120*** (0.000)
Treatment 3 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.305*** (0.000)	-0.095*** (0.000)
Treatment 4 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.280*** (0.000)	-0.070*** (0.000)
Differences from actual treatment effect	0.000	0.000	0.157	0.120	0.195
Observations	300000	300000	300000	300000	300000

Note: I simulate 10,000 observations across 30 periods, and randomly assign 20% to be treated in period 11. I define the outcome as taking a value of 0 for all units before period 11, and then vary the value based on different possible dynamic treatment effects. In column (1) treatment increases the outcome by 0.5 in the initial treatment period only. In column (2) treatment increases the outcome by 0.5 in each of the first 5 treatment periods. In column (3) treatment increases the outcome by 0.26 in all subsequent periods. In column (4) treatment increases the outcome by 0.5 in the first treatment period, but the effect *decreases* linearly over all subsequent periods. In column (5) treatment increases the outcome by 0.025 in the first treatment period, but the effect *increases* linearly over all subsequent periods. The evolution of the outcome over time is shown in [Figure C3](#). Patterns are similar with reversed signs if I simulate negative treatment effects. I estimate three treatment effect models for each scenario which make different assumptions about dynamic treatment effects. Panel A shows the results of event study estimates of the true effects for the first 5 treatment periods. Panel B shows the results of TWFE estimates that assume a transitory treatment that only affects the outcome in the initial treatment period. Panel C is similar to Panel B but includes four lagged treatment indicators, assuming treatment effects only persist for five periods. All regressions include period and unit fixed effects. SEs clustered at the unit level are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure C3: Simulation: evolution of outcome under different dynamic treatment effects

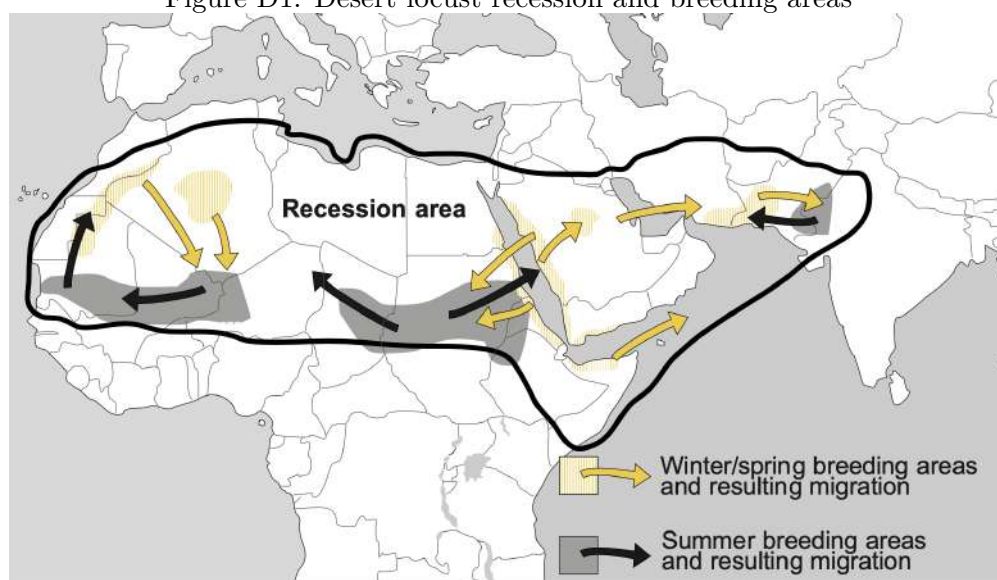


Note: This figure shows the evolution of the simulated outcomes as described in the notes to [Table C1](#).

D Desert locusts background

Desert locusts (*Schistocerca gregaria*) are a species of grasshopper always present in small numbers in desert ‘recession’ areas from Mauritania to India (Figure D1). They usually pose little threat to livelihoods but favorable climate conditions in breeding areas—periods of repeated rainfall and vegetation growth overlapping with the breeding cycle—can lead to exponential population growth. For example, the 2019-2021 locust upsurge persisted in large part because of repeated heavy precipitation out of season due to cyclones, prompting explosive reproduction (Cressman and Ferrand, 2021). The 2003-2005 upsurge was initiated by good rainfall over the summer of 2003 across four separate breeding areas. This was followed by two days of unusually heavy rains in October 2003 from Senegal to Morocco, after which environmental conditions were favorable for reproduction over the following 6 months (FAO and WMO 2016).

Figure D1: Desert locust recession and breeding areas



Source: Symmons and Cressman (2001). The recession area is the area in which desert locusts are most commonly found. The arrows indicate typical directions of downwind locust migration from breeding areas at different times of year.

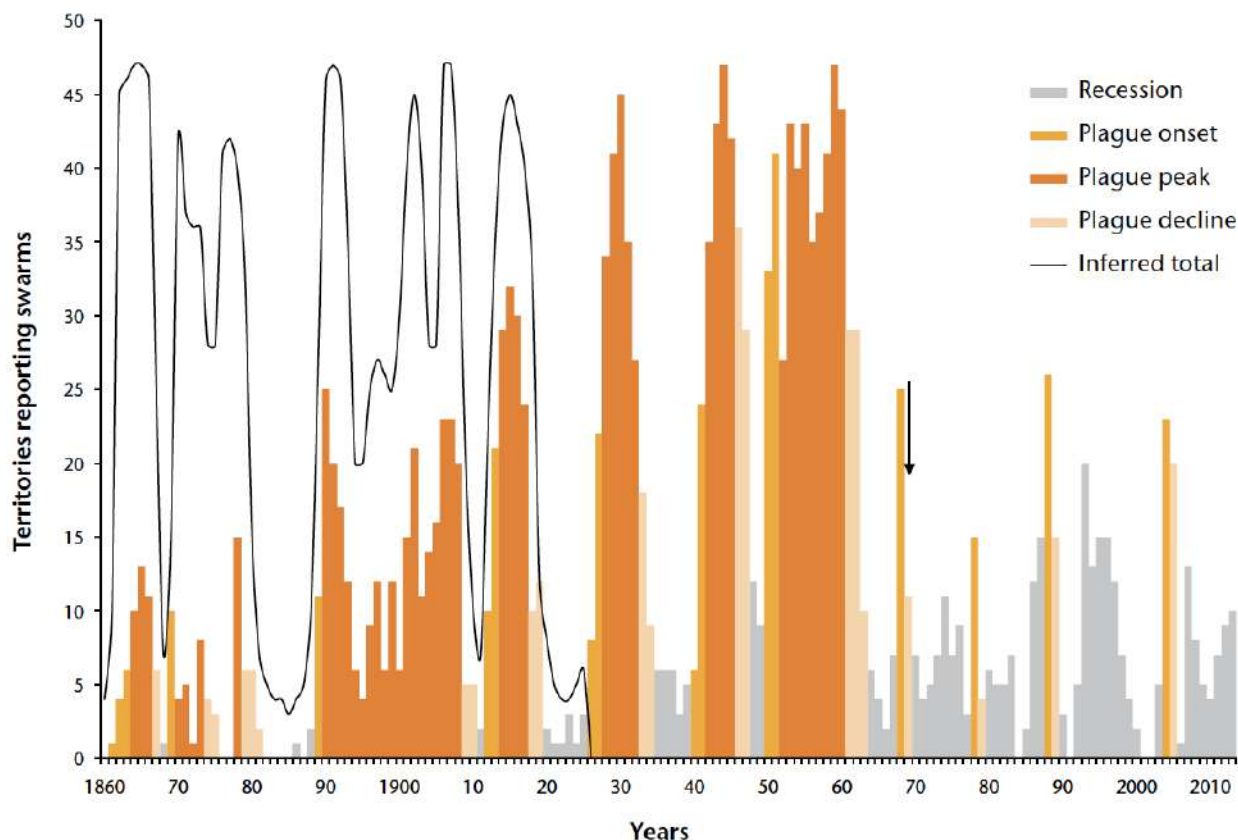
Unique among grasshopper species, after reaching a particular population density desert locusts undergo a process of ‘gregarization’ wherein they mature physically and form large bands or swarms which move as a cohesive unit (Symmons and Cressman, 2001). Locust bands may extend over several kilometers and alternate between roosting and marching, typically downwind (FAO and WMO 2016). In this paper, I focus my analysis on gregarious swarms of adult desert locusts and set aside observations of locusts in other phases.

Locust swarms form when bands of locusts remain highly concentrated when they reach the adult stage and become able to fly. Swarms vary in their density and extent (Symmons and Cressman, 2001). The average swarm includes around 50 locusts per m^2 with a range from 20-150, and can cover under 1 square kilometers to several hundred (Symmons and Cressman, 2001). About half of swarms exceed 50 km^2 in size (FAO and WMO 2016), meaning swarms typically include over a billion individuals.

The formation of swarms can lead to ‘outbreaks,’ where locusts spread out from their largely desert initial breeding areas. Locusts in swarms have increased appetites and accelerated reproductive cycles, and are thus particularly threatening to agriculture. The FAO distinguishes different levels of locust swarm activity (Symmons and Cressman, 2001). I use the terms ‘outbreak’ and ‘upsurge’ interchangeably to refer to any locust swarm activity. Few locust swarms are observed

outside of major outbreaks, as conditions favoring swarm formation tend to produce large swarms which reproduce and spread rapidly and are very difficult to control.

Figure D2: Desert locust observations by year



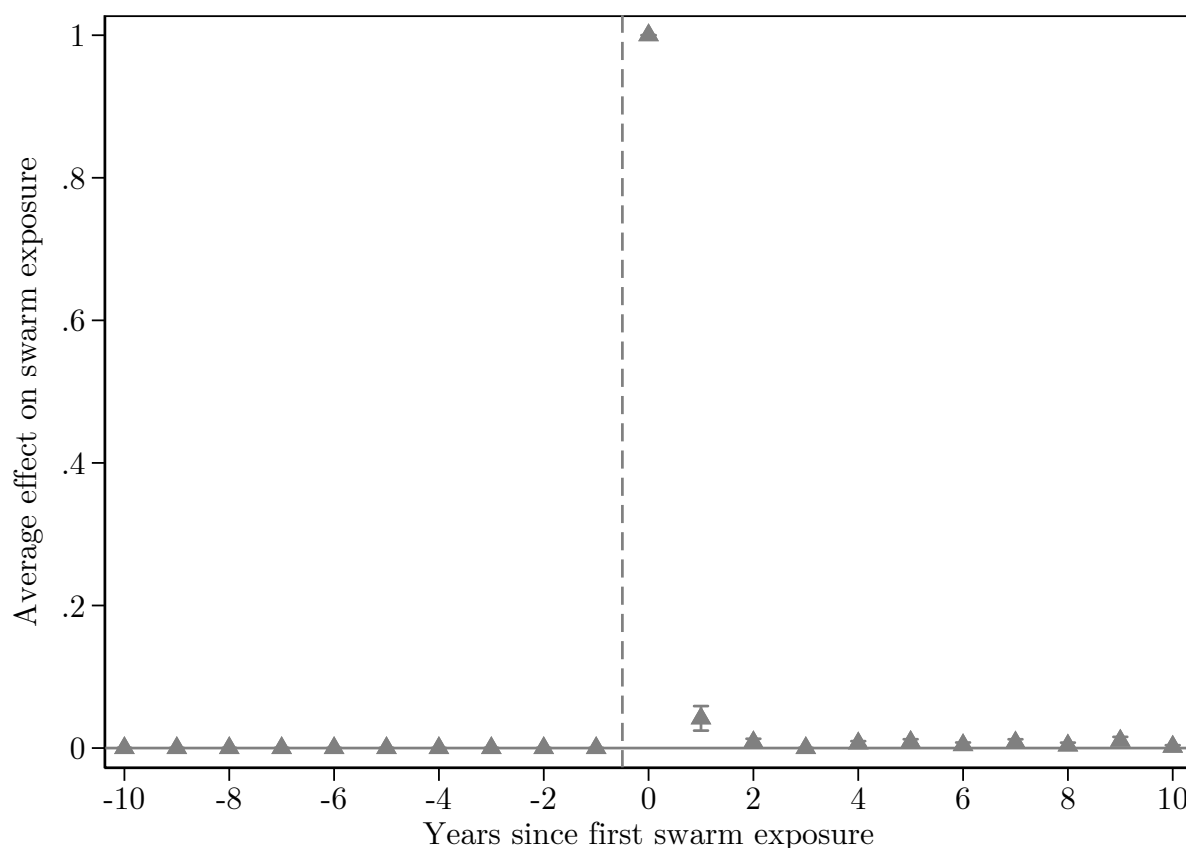
Source: Cressman and Stefanski (2016). Territories are generally individual countries.

The frequency of large-scale outbreaks has fallen since around the 1980s (Figure D2), in large part due to increases in coordinated preventive operations (Cressman and Stefanski, 2016). But climate change is expected to increase the risk of locust swarm formation and upsurges. Desert locusts can easily withstand elevated temperatures and the increased frequency of heavy rainfall events can create conditions conducive to population growth (Qiu, 2009; Youngblood et al., 2023).

As illustrated by Figure 1, locust swarms are not observed with any regularity over time or space. Desert locusts are migratory, moving on after consuming all available vegetation, and outside of outbreak periods are ultimately restricted to desert ‘recession’ areas. Unlike many other insect species, therefore, the arrival of a desert locust swarm does not signal a permanent change in local agricultural pest risk. Instead, the arrival of a swarm can be considered a locally and temporally concentrated natural disaster where all crops and pastureland are at risk (Hardeweg, 2001). Figure D3 illustrates how exposure to a locust swarm does not significantly affect the risk of exposure over the following years, consistent with exposure being a function of quasi-random variations in wind patterns and flight duration during swarm outbreaks. The only period in which cells exposed to locusts are more likely to experience a later locust swarm is in the period immediately after exposure, which reflects that many locust outbreaks last more than one year.

Locust swarms are migratory and fly downwind from a few hours after sunrise to an hour or so before sunset when they land and feed. Swarms do not always fly with prevailing winds and may wait for warmer winds. Small deviations in the positions of individual locusts in the swarm can also

Figure D3: Impact of swarm exposure on future exposure risk



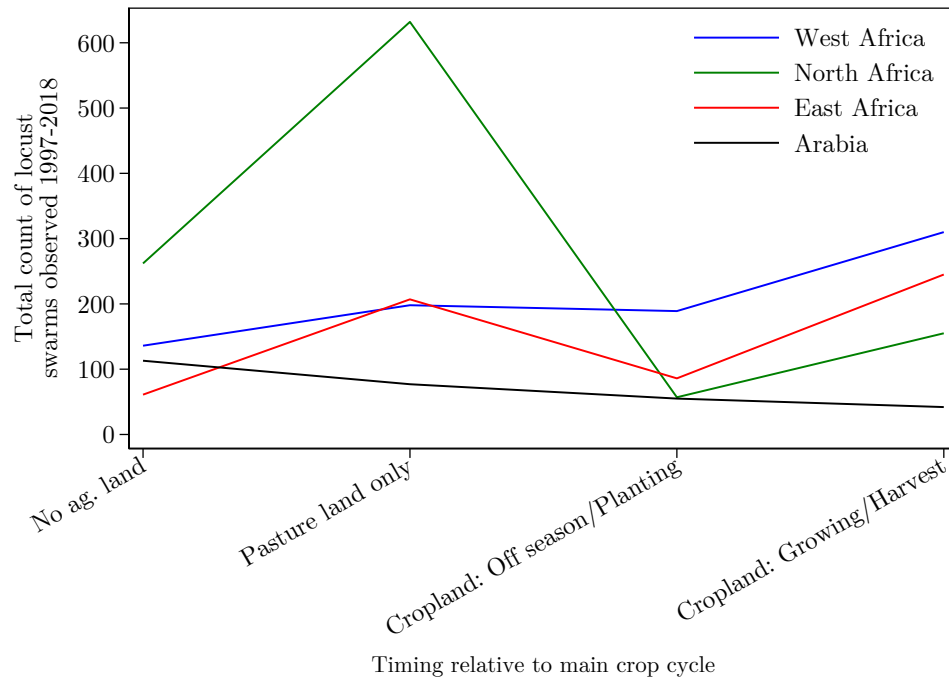
Note: The figure shows an event study of the impact of swarm exposure on the probability of having any locust swarm recorded in a subsequent year for the study period 1997-2018. By construction, none of the exposed areas had any locust swarm recorded in the years preceding their first exposure since 1990, and all had a locust swarm present in the year of exposure.

lead to changes in swarm flight trajectory, making their movements difficult to predict. Seasonal changes in these winds tend to bring locusts to seasonal breeding areas at times when rain and the presence of vegetation is most likely, allowing them to continue breeding (FAO and WMO 2016). Swarms are observed across different types of land cover and at different periods of the agricultural cycle, as shown in Figure D4.

Patterns in swarm movements lead to local variation in locust swarm exposure. After taking off, swarms fly for 9-10 hours rather than landing as soon as they encounter new vegetation. A swarm can easily move 100 km or more in a day even with minimal wind (Symmons and Cressman, 2001). Consequently, the flight path of a locust swarm will include both affected and unaffected areas, with the affected areas determined by largely by patterns of wind direction and speed over time from the initial swarm formation in breeding areas. Figure D5 illustrates the variation in areas affected by locust swarms over space around Mali. Swarm reports are densely clustered in the breeding areas in southern Mauritania where locust swarms reproduced in summer 2004. Outside of this area there is considerable variation in where swarms were reported, with distances between reported swarms over time consistent with typical flight distances.

These movement characteristics inform efforts to predict locust swarm movements, but these remain highly imprecise. The desert locust bulletins produced monthly by the FAO include forecasts of areas at risk of desert locust activity, but the areas described are quite large, often encompassing

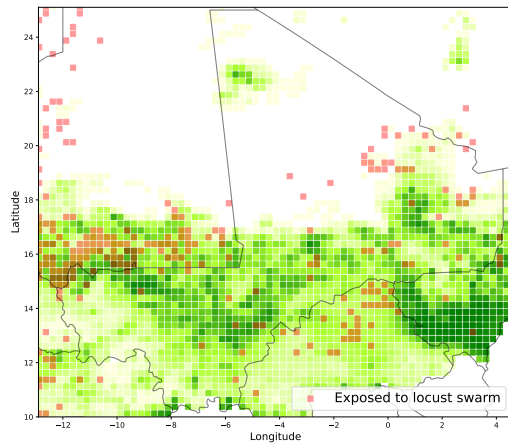
Figure D4: Variation in swarm timing and location during study period, by region



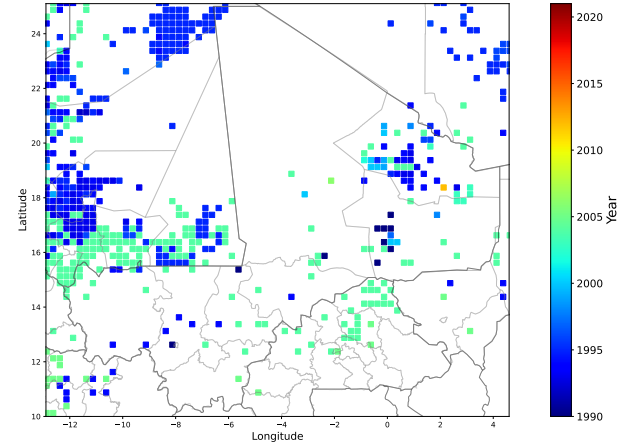
Note: The figure identifies the timing of swarm observations from 1997-2018 based on information on local crop calendars for major crops. This timing is not considered in cells with no cropland. These seasons are used to define timing of swarm exposure used in [Table 2](#).

Figure D5: Reports of locust swarms around Mali

A) Areas exposed to locust swarms since 1997



B) Year of first swarm exposure since 1990



Note: The figure illustrates the grid cells exposed to locusts swarms for the area around the country of Mali. Locust swarm reports are from the FAO Locust Watch database. Panel A overlays these reports on a map of the share of agricultural land area in each cell, while Panel B illustrates the timing of first exposure to locust swarms.

several countries in periods with increased swarms. While breeding regions and the broad areas at risk over different time periods can generally be predicted with some accuracy (Latchininsky, 2013; Samil et al., 2020; Zhang et al., 2019), predicting specific local variation in swarm presence remains a challenge due to the multiple factors influencing specific flight patterns (FAO and WMO 2016).

While desert locusts can exhibit some preferences over types of vegetation, avoiding woody

plants, dry vegetation, and some toxic plants in particular, swarms of locusts in their gregarious phase are less selective than solitary adults or nymphs (Despland, 2005; Despland and Simpson, 2005). The size of locust swarms and their polyphagous nature has led them to be considered the world's most dangerous and destructive migratory pest (Cressman et al., 2016; Lazar et al., 2016). A small swarm covering one square kilometer consumes as much food in one day as 35,000 people and the median swarm consumes 8 million kg of vegetation per day (FAO, 2023a), without preference for different types of crops (Lecoq, 2003).

The arrival of a locust swarm can therefore lead to the total destruction of local agricultural output (Showler, 2019). During the last locust upsurge in 2003-2005 in North and West Africa, 100, 90, and 85% losses on cereals, legumes, and pastures respectively were recorded, affecting more than 8 million people (Brader et al., 2006; Renier et al., 2015). Damages to crops alone were estimated at \$2.5 billion USD and \$450 million USD was required to bring an end to the upsurge (ASU, 2020). Over 25 million people in 23 countries were affected during the most recent 2019-2021 upsurge and damages were estimated to reach \$1.3 billion (Green, 2022), with control efforts—including treating over 2 million hectares with pesticides—estimated to have prevented over \$1 billion in damages (Newsom et al., 2021).

An important result of the local variation in locust swarm damages during outbreaks is that macro level impacts may be muted, since outbreaks occur in periods of positive rainfall shocks which tend to increase agricultural production in unaffected areas. Several studies find that impacts of locust outbreaks on national agricultural output and on prices are minimal, despite devastating losses in affected areas (Joffe, 2001; Krall and Herok, 1997; Showler, 2019; Thomson and Miers, 2002; Zhang et al., 2019). Chatterjee (2022) finds that wheat yields are 12% lower on average in Indian districts typically affected by desert locusts in years of locust outbreaks, in contrast to very large decreases in the specific areas exposed to locust swarms in those years.

Farmers have no proven effective recourse when faced with the arrival of a locust swarm, though activities such as setting fires, placing nets on crops, and making noise are commonly attempted. While these may slow damage they have little effect on locust population or total damages (Dobson, 2001; Hardeweg, 2001; Thomson and Miers, 2002). Desert locusts live 2-6 months and swarms continue breeding and migrating until dying out due to a combination of migration to unfavorable habitats, failure of seasonal rains in breeding areas, and control operations (Symmons and Cressman, 2001). The only current viable method of swarm control is direct air or ground spraying with pesticides (Cressman and Ferrand, 2021). These control operations do not prevent immediate agricultural destruction as they take some time to kill the targeted locusts, but will limit their spread. The 2003-2005 locust upsurge ended due to lack of rain and colder temperatures which slowed down the breeding cycle, combined with intensive ground and aerial spraying operations which treated over 130,000 km² at a cost of over US\$400 million (FAO and WMO 2016).

Desert locust control is most effective before locust populations surge, and the FAO manages an international network of early monitoring, warning, and prevention systems in support of this goal (Zhang et al., 2019). While improvements in desert locust management have been largely effective in reducing the frequency of outbreaks (as seen in Figure D2), many challenges remain. Desert locust breeding areas are widespread and often in remote or insecure areas. Small breeding groups are easy to miss by monitors, and swarms can migrate quickly. In addition, control operations are slow and costly, resources for monitoring and control are limited outside of upsurges, and the cross-country nature of the threat creates coordination issues. Insecurity may also limit locust control activities (Showler and Lecoq, 2021).